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Uncertainty Propagation and Dynamic Robust Risk Measures

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Abstract. We introduce a framework for quantifying propagation of uncertainty arising in a dynamic setting. Specifically, we define dynamic uncertainty sets designed explicitly for discrete stochastic processes over a finite time horizon. These dynamic uncertainty sets capture the uncertainty surrounding stochastic processes and models, accounting for factors such as distributional ambiguity. Examples of uncertainty sets include those induced by the Wasserstein distance and f-divergences. We further define dynamic robust risk measures as the supremum of all candidates' risks within the uncertainty set. In an axiomatic way, we discuss conditions on the uncertainty sets that lead to well-known properties of dynamic robust risk measures, such as convexity and coherence. Furthermore, we discuss the necessary and sufficient properties of dynamic uncertainty sets that lead to time-consistencies of dynamic robust risk measures. We find that uncertainty sets stemming from f-divergences lead to strong time-consistency whereas the Wasserstein distance results in a new time-consistent notion of weak recursiveness. Moreover, we show that a dynamic robust risk measure is strong time-consistent or weak recursive if and only if it admits a recursive representation of one-step conditional robust risk measures arising from static uncertainty sets.

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Keywords: dynamic risk measures • time-consistency • distributional uncertainty • Wasserstein distance

1. Introduction

As uncertainty prevents perfect information from being attained, decision makers are confronted with the consequences of their risk assessments made under partial information. Incorporating model misspecification and Knightian uncertainty into dynamic decision making, thus robustifying one's decisions, has been studied in various fields, including economics (Epstein and Schneider [23], Maccheroni et al. [39], Siniscalchi [51]), mathematical finance (Bielecki et al. [11], Neufeld et al. [42]), and risk management (Acciaio et al. [3]). Many circumstances require sequential decisions, where risk assessments are made over a finite time horizon and are based on the flow of information. Importantly, these decisions need to be time-consistent (t.c.) and account for the propagation of uncertainty. Although the theory of t.c. dynamic risk measures is growing (Artzner et al. [5], Bielecki et al. [10], Cheridito and Kupper [16], Cheridito et al. [18], Feinstein and Rudloff [25], Jobert and Rogers [34], Riedel [46]), the incorporation of dynamic uncertainty into dynamic risk measures is only little explored. In the economic literature, the theory of recursive multiple-priors and variational preferences is closest to our work. While working with a recursive notion of time-consistency, most works focus on dynamic utility, event trees, and uncertainty sets that are fixed throughout time; see, for example, Epstein and Schneider [23], Maccheroni et al. [39], Siniscalchi [51], and references therein. Here, we work with dynamic risk measures, stochastic processes, and uncertainty that may change over time. Another related area of research is reinforcement learning (RL). Although there are recent works in the field of distributional robust RL in the context of Markov decision processes that account for uncertainty in the underlying processes, they typically maximise expected reward. The distributional robust risk-aware RL literature, which instead of expected rewards optimises risk measures, mostly deals with static uncertainty and static risk measures. Indicatively, see, for example, Abdullah et al. [1] and Smirnova et al. [52], who model uncertainty on the transition probabilities, and Jaimungal et al. [33], who consider uncertainty only at terminal time.

In this work, we propose an axiomatic framework for quantifying uncertainty of discrete time stochastic processes and tie them to robustified dynamic risk measures. Specifically, we introduce *dynamic uncertainty sets* consisting of a family of time-t uncertainty sets. Each time-t uncertainty set is a collection of $\mathcal{F}_t$-measurable random variables summarising the uncertainty of the entire stochastic process at time t. The dynamic uncertainty sets may vary with each stochastic process, as the uncertainty of two processes may differ, even if they share the same law. This general framework includes, to the authors' knowledge, all uncertainty sets encountered in the literature, from moment constraints, f-divergences, seminorms, the popular (adapted) Wasserstein distance, and ambiguity in a base probability.

Equipped with a strong t.c. dynamic risk measure and a dynamic uncertainty set, we define *dynamic robust risk measures* as sequences of conditional robust risk measures, by taking the supremum of all risks in the uncertainty set. We then proceed by studying conditions on the dynamic uncertainty set that lead to well-known properties of dynamic robust risk measures such as convexity and coherence. Crucial to the dynamical framework are notions of time-consistencies, of which many have been introduced and studied in the literature. The most common is strong time-consistency—often also referred to as recursiveness—leading to a dynamic programming principle (Cheng and Jaimungal [15], Cheridito et al. [18], Ruszczyński [48]). Although the majority of works assume normalisation of dynamic risk measures, in a robust setting, incorporating uncertainty does not necessarily result in the robustified dynamic risk measures being normalised. Indeed, an important subject of debate is whether the value of zero—or more generally an $\mathcal{F}_{t-1}$-measurable random variable—contains uncertainty—at time t. We find that uncertainty sets induced by the f-divergence are normalised whereas those generated by the Wasserstein distance or a norm, with, for example, a constant tolerance distance, are not. Consequently, we introduce a new concept of weak recursiveness to account for uncertainty sets that are not normalised.

One of the manuscript's key theorems generalises results from the seminal works of Cheridito et al. [18] and Ruszczyński [48] to account for dynamic uncertainty. Specifically, we show that a dynamic robust risk measure is strong t.c. or weak recursive if and only if it admits a recursive representation of one-step robust risk measures. Furthermore, these one-step robust risk measures are characterised by dynamic uncertainty sets which possess the property of static. Static uncertainty sets arise in one-period settings and only account for uncertainty at the current time. Thus, we show that when working with strong t.c. or weak recursive dynamic robust risk measures, it is enough to consider the simpler subclass of static uncertainty sets.

The paper is organised as follows. Section 2 presents the framework of dynamic uncertainty sets, discusses their properties, and provides examples. In Section 3, we define dynamic robust risk measures and show, in an axiomatic way, sufficient and necessary conditions on the dynamic uncertainty set that provide properties of dynamic robust risk measures. Section 4.1 introduces different notions of time-consistencies of dynamic uncertainty sets and connects them to time-consistencies of dynamic robust risk measures. Furthermore, in Section 4.2 we prove the recursive representation of strong t.c. and weakly recursive dynamic robust risk measures. We conclude with examples of dynamic robust risk measures in Section 5. Proofs not included in the main body are delegated to the appendix.

2. Uncertainty in a Dynamic Setting

This section proposes a framework for axiomatically quantifying uncertainty in a dynamic setting.

2.1. Preliminaries and Notation

Consider a finite time horizon $T \in \mathbb{N}$, and a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \{\mathcal{F}_t\}_{t\in\{0,1,\dots,T\}})$. For $t \in \{0,1,\dots,T\}$, we denote by $L_t^\infty := L^\infty(\Omega, \mathcal{F}_t, \mathbb{P})$ the space of $\mathcal{F}_t$-measurable bounded random variables and set $L_{0,T}^\infty := L_0^\infty \times \cdots \times L_T^\infty$. This setup allows for the representation of one-dimensional stochastic processes $X \in L_{0,T}^\infty$, $X := \{X_0, X_1, \dots, X_T\}$, where each $X_t \in L_t^\infty$ represents a (discounted) loss at time t, and if not otherwise stated we set for simplicity $X_0 = 0$. We further define the spaces $L_{t,s}^\infty := 0 \times \cdots \times 0 \times L_t^\infty \times \cdots \times L_s^\infty \times 0 \times \cdots \times 0$, for all $0 \le t < s \le T$. For $X \in L_{0,T}^\infty$, we denote by $X_{t:s}$, $0 \le t < s \le T$ its projection onto the space $L_{t,s}^\infty$, that is, $X_{t:s} := \{0, \dots, 0, X_t, \dots, X_s, 0, \dots, 0\}$. Whenever there is no confusion, we omit the zeroes; that is, we write $X_{t:s} = \{X_t, \dots, X_s\}$. Thus, for $X_{s:T} \in L_{s,T}^\infty$ and Z_t with $t < s$, we may write $X_{s:T} + Z_t = (\underbrace{0, \dots, 0}_{t}, Z_t, \underbrace{0, \dots, 0}_{s-t-1}, X_s, \dots, X_T)$. We further define the supremum norm on the spaces $L_{t,s}^\infty$ as

$$\|X_{t:s}\|_{t,s} := \operatorname{ess\,inf}\left\{ m \in L_t^\infty : \sup_{t \le i \le s} |X_i| \le m \right\}. \tag{1}$$

If not otherwise stated, all equalities and inequalities between random vectors are component-wise and in a $\mathbb{P}$-almost sure (a.s.) sense. Central to the exposition are set-valued functionals. To clarify the notation, we recall the

sum of sets,

$$A + B := \{X + Y \in L^\infty_{0,T} : X \in A,\, Y \in B\}, \quad \text{where} \quad A, B \subseteq L^\infty_{0,T}.$$

By abuse of notation, we may denote sets consisting of a singleton by its element, that is, $Z := \{Z\} \subset L^\infty_{0,T}$. We further recall the multiplication of a set $A \subseteq L^\infty_t$ with a $\mathcal{F}_t$-measurable random variable $\lambda \in L^\infty_t$ as

$$\lambda A := \{\lambda X \in L^\infty_t : X \in A\},$$

and denote the complement of a set $A \subseteq L^\infty_t$ by $A^\complement := \{X \in L^\infty_t : X \notin A\}$.

2.2. Dynamic Uncertainty Sets

In this section, we introduce the notion of *dynamic uncertainty sets*, which quantifies uncertainty around stochastic processes. For this, we define the notation $\mathcal{T} := \{0, \dots, T-1\}$ and $\overline{\mathcal{T}} := \{1, \dots, T\}$.

Definition 1 (Dynamic Uncertainty Set). A dynamic uncertainty set $\mathfrak{u} := \{u_t\}_{t \in \overline{\mathcal{T}}}$ is a sequence of time-t uncertainty sets $\{u_t\}_{t \in \overline{\mathcal{T}}}$, where for each $t \in \overline{\mathcal{T}}$, the time-$t$ uncertainty set u_t is a mapping $u_t : L^\infty_{0,T} \to 2^{L^\infty_t}$.

A time-t uncertainty set u_t is thus a set function mapping a stochastic process $X_{0:T}$ to a subset of $\mathcal{F}_t$-measurable random variables and could thus include uncertainty of the entire time horizon of the processes. If an uncertainty set is evaluated on a projection of a stochastic process, for example, on $X_{t:s}$, $t, s \in \overline{\mathcal{T}}$ with $t \le s$, then we simply write $u_t(X_{t:s}) := u_t(0, \dots, 0, X_t, \dots, X_s, 0, \dots, 0) \subseteq L^\infty_t$.

In the context of, for example, financial losses represented by $X_{t:T}$, uncertainty at time t may arise as the agent encounters ambiguity whether $X_{t:T}$ accurately models the true loss. In such instances, it becomes prudent for the agent to consider a set of alternative stochastic processes. These alternative stochastic processes are losses that are considered "close" to $X_{t:T}$ or share common attributes such as similar distributional features. The uncertainty at time t, that is, $u_t(X_{t:T})$, may then be viewed as the projection of the space of alternative processes onto time t. In Section 2.3 we present several examples of uncertainty sets from the literature adapted to the dynamic setting.

Properties 2.1. *A time-t uncertainty set u_t may satisfy the following:*

i. ***Proper:*** *Nonempty and bounded from above*[1] *for all $X_{t:T} \in L^\infty_{t,T}$.*

ii. ***Normalisation:*** $u_t(0) := u_t(0, \dots, 0) = \{0\}$.

iii. ***Order preservation:*** *Let $X_{t:T} \le Y_{t:T}$ with $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$. Then for each $Z \in u_t(X_{t:T})$ there exists a $W \in u_t(Y_{t:T})$ such that $Z \le W$.*

iv. ***Monotonicity:*** *$X_{t:T} \le Y_{t:T}$ implies that $u_t(X_{t:T}) \subseteq u_t(Y_{t:T})$, for all $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$.*

v. ***Translation invariance:*** *$u_t(X_{t:T} + Z_s) = u_t(X_{t:T}) + Z_s$ for all $X_{t:T} \in L^\infty_{t,T}$ and $Z_s \in L^\infty_s$ with $s < t$.*[2]

vi. ***Static:*** *$u_t(X_{t:T}) = u_t(X_t)$, for all $X_{t:T} \in L^\infty_{t,T}$.*

vii. ***Locality:*** *$u_t(1_B X_{t:T} + 1_{B^\complement} Y_{t:T}) = 1_B\, u_t(X_{t:T}) + 1_{B^\complement}\, u_t(Y_{t:T})$ for all $B \subseteq L^\infty_t$ and $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$.*

viii. ***Positive homogeneity:*** *$u_t(\lambda X_{t:T}) = \lambda\, u_t(X_{t:T}) + 1_{\lambda=0}\, u_t(0)$ for all $0 \le \lambda \in L^\infty_{t-1}$ and $X_{t:T} \in L^\infty_{t,T}$.*

ix. ***Star-shapedness:*** *$u_t(\lambda X_{t:T}) \subseteq \lambda\, u_t(X_{t:T}) + 1_{\lambda=0}\, u_t(0)$ for all $\lambda \in L^\infty_{t-1}$ with $0 \le \lambda \le 1$ and $X_{t:T} \in L^\infty_{t,T}$.*

We say a dynamic uncertainty set $\mathfrak{u} = \{u_t\}_{t \in \overline{\mathcal{T}}}$ satisfies one of the above properties if u_t satisfies it for all $t \in \overline{\mathcal{T}}$. Clearly, a time-$t$ uncertainty set should be nonempty and bounded. Normalisation pertains to whether zero is ambiguous, a property that may or may not be desired. In Section 2.3 we see that uncertainty sets induced by norms are in general not normalised whereas those by f-divergences are. In the context of hedging, an optimal strategy may attain a liability of $X_{0:T} = 0$. However, the underlying model may not capture all aspects, such as market liquidity and legal and political risk, which can be accounted for through a nonnormalised uncertainty set. Monotonicity, which states that the uncertainty set of a dominating stochastic process is larger than the original process, is a strong property, however important for the exposition. The weaker notion of order preservation (implied by monotonicity) states that any element in $u_t(X_{t:T})$ is a.s. dominated by an element in the uncertainty set of the dominating process $Y_{t:T}$. Translation invariance means that known information does not change the uncertainty set and is useful for incorporating prior information, such as Bayesian updating. For example, if $\mathfrak{u}$ is normalised and translation invariant, then any $\mathcal{F}_{t-1}$-measurable random variable does not exhibit uncertainty at time t. A static time-t uncertainty set only accounts for uncertainty around X_t. Indeed, if $\mathfrak{u}$ is static, then it satisfies $u_t(X_t) = u_t(X_t + Y_{t+1:T})$ for any $Y_{t+1:T} \in L^\infty_{t+1,T}$; thus, the uncertainty set is indifferent about the future of the process. Positive homogeneity implies that the uncertainty set scales with the size of the loss, whereas star-shapedness means that it increases with size.

Lemma 1. *Let u_t be a time-t uncertainty set. Then,*

i. *u_t is normalised and local if and only if $u_t(1_B X_{t:T}) = 1_B\, u_t(X_{t:T})$ for all $B \in \mathcal{F}_{t-1}$ and $X_{t:T} \in L^\infty_{t,T}$;*

ii. *if u_t is positive homogeneous, then u_t is local.*

The proof is delegated to the appendix.

2.3. Examples of Dynamic Uncertainty Sets

In the static setting, there is a plethora of uncertainty sets considered in the literature, including uncertainty sets defined via, for example, mixtures of distributions (Zhu and Fukushima [56]), moment constraints (Hürlimann [31], Natarajan et al. [41]), divergence constraints (Ben-Tal et al. [7], Calafiore [14], Jaimungal et al. [33], Luo and Mehrotra [38], Wang et al. [55]), and combinations of moment and divergence constraints (Bernard et al. [8]). Furthermore, in the study of convex risk measures, robustness is closely associated with the underlying probability measure, with Artzner et al. [4] and Delbaen [20] emphasising uncertainty sets centred around a probability measure. However, a limitation of the latter perspective is that a random variable's uncertainty only depends on its distribution. The proposed approach goes beyond by considering uncertainty sets of random variables rather than probability measures.

In this section we focus on examples of uncertainty sets that are constructed as "balls" around a reference distribution with radius given by a tolerance distance. In the dynamic setting, the time-t reference distribution could be the distribution of $X_{t:T}$ conditional on $\mathcal{F}_{t-1}$, to, for example, account for uncertainty in transition probabilities. The time-t tolerance distance may depend on X_t and encompass information available at time t such as market cyclicality, trading volume, and market liquidity. The first set of examples pertains to uncertainty sets induced by seminorms or norms on the spaces of random variables and stochastic processes, such as the total variation norm, p-norm, Hölder norm, and supremum norm; see, for example, Gotoh and Takeda [29] and Gotoh et al. [30] for the static setting. Second, we discuss uncertainty sets induced by the (adapted) Wasserstein distance. Uncertainty sets characterised by the Wasserstein distance in the static case have become popular; indicatively, see Gao and Kleywegt [28], Jaimungal et al. [33], Mohajerin Esfahani and Kuhn [40], Pesenti and Jaimungal [43], Pflug and Wozabal [45], and Backhoff-Veraguas et al. [6] for the adapted Wasserstein distance. We also refer to Blanchet and Murthy [13] and Pesenti and Vanduffel [44] for uncertainty sets based on optimal transport between probability measures. The last set of examples considers distances on the space of probability distributions, and in particular, we consider dynamic uncertainty sets induced by f-divergences and the Kullback-Leibler (KL) divergence; see, for example, Jaimungal et al. [32] and Kroell et al. [35] for the KL divergences on the space of stochastic processes. Table 1 summarises the properties of the dynamic uncertainty sets.

Example 1 (Seminorm on Random Variables). Consider the dynamic uncertainty set given by

$$u_t^{\|\cdot\|}(X_t) := \{Y \in L_t^\infty \ : \|X_t - Y\| \le \varepsilon_{X_t}\}, \quad t \in \overline{\mathcal{T}}, \tag{2}$$

where $\|\cdot\| : L_t^\infty \to L_{t-1}^\infty$ is a (random) seminorm and $\varepsilon_{X_t} \ge 0$, $\varepsilon_{X_t} \in L_{t-1}^\infty$ is a tolerance distance. The choice of tolerance distance includes, for example, $\varepsilon_{X_t} = \varepsilon$, $\varepsilon_{X_t} = (T-t)\varepsilon$, $\varepsilon_{X_t} = \varepsilon \operatorname{var}(X_t|X_{t-1})$, with $\varepsilon \in \mathbb{R}$ and where $\operatorname{var}(\cdot)$ denote the conditional variance.

This time-t uncertainty set is proper and static. It is normalised if and only if $\|\cdot\|$ is a norm and satisfies $\varepsilon_0 = 0$, that is, $\varepsilon_{X_t} = 0$ whenever $X_t = 0$. If $\varepsilon_{X_t} \le \varepsilon_{Y_t}$ for all $X_t \le Y_t \in L_t^\infty$, then the uncertainty set is order preserving. It is translation invariant, if $\varepsilon_{X_t+Y_{t-1}} = \varepsilon_{X_t}$ for all $X_t \in L_t^\infty$ and $Y_{t-1} \in L_{t-1}^\infty$. If $\varepsilon_{\lambda X_t} = \lambda \varepsilon_{X_t}$, for all $X_t \in L_t^\infty$ and $0 \le \lambda \in L_{t-1}^\infty$, then the uncertainty set is positive homogeneous, and consequently local. Similarly, if $\varepsilon_{\lambda X_t} \le \lambda \varepsilon_{X_t}$ for all $X_t \in L_t^\infty$ and $0 \le \lambda \in L_{t-1}^\infty$, then the uncertainty set is star-shaped. Moreover, if $\varepsilon_{X_t+Y_t} \ge \varepsilon_{X_t} + \varepsilon_{Y_t}$ for all $X_t, Y_t \in L_t^\infty$, then $u_t(X_t) + u_t(Y_t) \subseteq u_t(X_t + Y_t)$.

Table 1. Examples of dynamic uncertainty sets discussed in Section 2.3 and their properties.

Properties	Seminorm	Wasserstein	Probability	Cond. KL
	Equation (2), $u_t^{\|\cdot\|}$	Equation (6), u_t^W	Equation (7), $u_t^{\mathcal{Q}}$	Equation (8), u_t^{KL}
Proper	√	√	√	√
Normalised	Norm, $\varepsilon_0 = 0$	$\varepsilon_0 = 0$	√	√
Order preserving	$\varepsilon_{X_t} \le \varepsilon_{Y_t}$	$\varepsilon_{X_t} \le \varepsilon_{Y_t}$	√	$\varepsilon_{X_t} \le \varepsilon_{Y_t}$
Translation invariant	$\varepsilon_{X_t+Y_t} = \varepsilon_{X_t}$	$\varepsilon_{X_t+Y_t} = \varepsilon_{X_t}$	√	√
Static	√	√	√	×
Local	$\varepsilon_{\lambda X_t} = \lambda\, \varepsilon_{X_t}$	$\varepsilon_{\lambda X_t} = \lambda\, \varepsilon_{X_t}$	√	√
Positive homogeneous	$\varepsilon_{\lambda X_t} = \lambda\, \varepsilon_{X_t}$	$\varepsilon_{\lambda X_t} = \lambda\, \varepsilon_{X_t}$	√	√

Note. Cond., conditional.

For the special case when the tolerance distance $\varepsilon^\dagger \in L^\infty_{t-1}$ satisfies $\varepsilon^\dagger = \varepsilon^\dagger_{X_t}$ for all $X_t \in L^\infty_t$, then the uncertainty set (2) reduces to

$$u_t^{\|\cdot\|}(X_t) = \{Y + X_t \in L^\infty_t : \|Y\| \le \varepsilon^\dagger\} = u_t^{\|\cdot\|}(0) + X_t, \tag{3}$$

which implies that the uncertainty set of X_t is entirely described by the uncertainty around the origin.

Example 2 (Seminorm on Stochastic Processes). Consider the dynamic uncertainty set with a seminorm $\|\cdot\| : L^\infty_{t,T} \to L^\infty_t$ and tolerance distance $0 < \varepsilon_{X_{t:T}} \in L^\infty_t$, given by

$$u_t(X_{t:T}) := \{Y \in L^\infty_t : \|X_{t:T} - Y\| \le \varepsilon_{X_{t:T}}\}, \quad t \in \overline{\mathcal{T}}. \tag{4}$$

If the norm is the supremum norm given in (1), that is, $\|\cdot\| := \|\cdot\|_{t:T}$, then the uncertainty set becomes, for $t < T$,

$$u_t(X_{t:T}) = \{Y \in L^\infty_t : \max\{\|X_{t+1:T}\|_{t+1:T}, \operatorname{ess\,sup}\{X_t - Y\}\} \le \varepsilon_{X_{t:T}}\}.$$

If further $\varepsilon_{X_{t:T}} < \|X_{t+1:T}\|_{t+1:T}$, then $u_t(X_{t:T}) = \emptyset$, and if $\varepsilon_{X_{t:T}} \ge \|X_{t+1:T}\|_{t+1:T}$, then u_t simplifies to

$$u_t(X_{t:T}) := \{Y \in L^\infty_t : \|X_t - Y\|_t \le \varepsilon_{X_{t:T}}\}. \tag{5}$$

When $\|\cdot\|$ is the sum of norms, that is, $\|Y_{t:T} - X_{t:T}\| := \sum_{s=t}^T \|X_s - Y_s\|_{s:s}$, and $\varepsilon_{X_{t:T}} < \varepsilon_{X_t} + \|X_{t+1:T}\|$, for some $\varepsilon_{X_t} > 0$, then the uncertainty set is empty. For $\varepsilon_{X_{t:T}} = \varepsilon_{X_t} + \|X_{t+1:T}\|$, u_t becomes identical to the one in Example 1.

Alternatively, we can define the uncertainty set as the L^∞_t-projection of stochastic processes, that is,

$$u_t(X_{t:T}) := \{Y_t \in L^\infty_t : \|X_{t:T} - Y_{t:T}\| \le \varepsilon_{X_{t:T}}\}.$$

For the Hölder and total variation norm, this uncertainty set is equal to the entire space of $\mathcal{F}_t$-measurable random variables, that is, $u_t(X_{t:T}) = L^\infty_t$, and thus is not proper. For the p-norm, the uncertainty set reduces to (5) with the p-norm on L^∞_t. Thus, many uncertainty sets based on norms for processes lead to pathological (not proper) uncertainty sets or reduce to those in Example 1.

Example 3 (Wasserstein Uncertainty). Consider the time-t uncertainty set induced by the p-conditional Wasserstein distance, $p \ge 1$,

$$u_t^W(X_t) := \left\{ Y \in L^\infty_t : \int_0^1 |F^{-1}_{Y|\mathcal{F}_{t-1}}(\alpha) - F^{-1}_{X_t|\mathcal{F}_{t-1}}(\alpha)|^p d\alpha \le \varepsilon^p_{X_t} \right\}, \tag{6}$$

where $F^{-1}_{Y|\mathcal{F}_{t-1}}$ denotes the conditional left-quantile function of Y given $\mathcal{F}_{t-1}$; see, for example, de Castro et al. [19] for a definition. This uncertainty set is order preserving, translation invariant, positive homogeneous, and normalised if ε_{X_t} satisfies the respective properties as in Example 1. Uncertainty sets of this type are studied in Gao and Kleywegt [28], Luo and Mehrotra [37], Luo and Mehrotra [38], and Mohajerin Esfahani and Kuhn [40], where ε_{X_t} is not a function of X_t and therefore, the uncertainty sets are not normalised.

Alternatively, the Wasserstein distance for stochastic processes gives rise to

$$u_t(X_{t:T}) := \{Y_t \in L^\infty_t : W(X_{t:T}, Y_{t:T}) \le \varepsilon_{X_{t:T}}\}, \quad \text{with}$$
$$W(X_{t:T}, Y_{t:T}) := \inf\{(\|X'_{t:T} - Y'_{t:T}\|^p) : F_{X'_{t:T}|\mathcal{F}_{t-1}} = F_{X_{t:T}|\mathcal{F}_{t-1}}, F_{Y'_{t:T}|\mathcal{F}_{t-1}} = F_{Y_{t:T}|\mathcal{F}_{t-1}}\},$$

where the infimum is taken over all joint distributions $(X'_{t:T}, Y'_{t:T})$ with given conditional marginals, $F_{Y|\mathcal{F}_{t-1}}$ denotes the conditional cumulative distribution function (cdf) of Y given $\mathcal{F}_{t-1}$, and $\|\cdot\|^p$ is the p-norm. This uncertainty set has the same properties as the one in (6). Because $\|X'_t - Y'_t\|^p \le \|X'_{t:T} - Y'_{t:T}\|^p \le \|X'_t - Y'_t\|^p + \|X'_{t+1:T} - Y'_{t+1:T}\|^p$ and we can choose $Y'_{t+1:T} = X'_{t+1:T}$, it follows that $\|X'_t - Y'_t\|^p = \|X'_{t:T} - Y'_{t:T}\|^p$. Hence, if $\varepsilon_{X_{t:T}} = \varepsilon_{X_t}$, this uncertainty set reduces to (6). The same holds for the adapted Wasserstein distance.

Example 4 (Uncertainty on the Probability). Uncertainty may arise from the underlying probability measure, such as in the context of model risk. For this, we denote by F_X the cdf of X under the base probability measure $\mathbb{P}$. Further, let $\mathcal{Q}$ be a set of probability measures that are absolutely continuous with respect to (w.r.t.) $\mathbb{P}$ and consider the uncertainty set

$$u_t^{\mathcal{Q}}(X_t) := \{Y \in L^\infty_t : F_{Y|\mathcal{F}_{t-1}} = F^{\mathbb{Q}}_{X_t|\mathcal{F}_{t-1}}, \text{ for some } \mathbb{Q} \in \mathcal{Q}\}, \tag{7}$$

where $F^{\mathbb{Q}}_X$ is the cdf of X under $\mathbb{Q}$. Then $u_t^{\mathcal{Q}}$ is proper, normalised, order preserving, translation invariant, local, and positive homogeneous. Such an uncertainty set is proposed in Blanchet and Murthy [13], where $\mathcal{Q}$ is based on

optimal transport between probability measures. When $\mathcal{Q}$ contains probability measures that are not absolutely continuous w.r.t. $\mathbb{P}$, the above properties of $u_t^{\mathcal{Q}}$ may not hold. To illustrate, let $\mathbb{P}, \mathbb{Q} \in \mathcal{Q}$ where $\mathbb{Q}$ is not absolutely continuous w.r.t. $\mathbb{P}$. Then there exists a set A such that $\mathbb{P}(A) = 0 < \mathbb{Q}(A)$. Consequently, as $1_A = 0$ $\mathbb{P}$-a.s., we have $u_1^{\mathcal{Q}}(0) = u_1^{\mathcal{Q}}(1_A) \supset \{0\}$, where the inclusion is strict.

Example 5 (Uncertainty Induced by Divergences). Let D_t be a function mapping cdfs to L_{t-1}^{∞}, that is, $(F, G) \longmapsto D_t(F, G) \in L_{t-1}^{\infty}$, and consider its induced uncertainty set

$$u_t^D(X_{t:T}) := \{Y_t \in L_t^{\infty} : D_t(F_{Y_{t:T}}, F_{X_{t:T}}) \leq \varepsilon_{X_{t:T}}\}. \tag{8}$$

Examples of D_t include f-divergences and, in particular, the KL divergence. The uncertainty set u_t^D is normalised, whenever the divergence of distributions with differing support is equal to infinity—which is the case for f-divergences. It is translation invariant, if $D_t(F_{Y_{t:T}}, F_{X_{t:T}+c}) = D_t(F_{Y_{t:T}-c}, F_{X_{t:T}})$, for all $c \in L_{t-1}^{\infty}$, and $\varepsilon_{X_{t:T}} = \varepsilon_{X_{t:T}+c}$. It is positive homogeneous, if $D_t(F_{Y_{t:T}}, F_{\lambda X_{t:T}}) = D_t(F_{\frac{Y_{t:T}}{\lambda}}, F_{X_{t:T}})$ and $\varepsilon_{X_{t:T}} = \varepsilon_{\lambda X_{t:T}}$, for all $0 < \lambda \in L_{t-1}^{\infty}$. Uncertainty sets induced by conditional f-divergences, in particular, the conditional KL divergence, satisfy the above. Furthermore, uncertainty sets induced by f-divergences are order preserving if $\varepsilon_{X_{t:T}} \leq \varepsilon_{Y_{t:T}}$ for all $X_{t:T} \leq Y_{t:T}$.

If $\mathcal{Q}$ is given by $\mathcal{Q} := \{\mathbb{Q} : d_t(F_{X_{t:T}}, F_{X_{t:T}}^{\mathbb{Q}}) \leq \varepsilon\}$, then the uncertainty sets in Equations (7) and (8) coincide.

3. Dynamic Robust Risk Measure

Next, we propose a class of dynamic robust risk measures that incorporates the dynamic uncertainty sets introduced in the last section. Specifically, we are interested in robustifying strong t.c. dynamic risk measures, which are normalised, monotone, and translation invariant, by taking at each time point the worst-case value of the dynamic risk measure. Strong t.c. dynamic risk measures that are normalised, monotone, and translation invariant are studied extensively, as they allow for a recursive representation (see Theorem 1), which in many settings leads to a dynamic programming principle (Ruszczyński [48]) which allows solving multistep optimisation problems (Shapiro et al. [50]). When robustifying a dynamic risk measure, however, some of their characteristics may get lost; thus, this section studies the necessary and sufficient requirements on dynamic uncertainty sets that preserve the properties of dynamic robust risk measures.

3.1. Definition and Properties

We first recall the definitions and properties of conditional and dynamic risk measures and refer the reader to Cheridito et al. [17], Frittelli and Scandolo [26], Föllmer and Schied [27], and Ruszczyński and Shapiro [49] for discussions and interpretations, as well as Laeven et al. [36] for star-shaped dynamic risk measures.

As uncertainty may change over time, we consider the dynamic risk of the entire process rather than the total loss amount at terminal time. In particular, when incorporating uncertainty, we do not assume that the uncertainty sets respect translation invariance; hence, working with the entire process becomes necessary.

Definition 2 (Dynamic Risk Measure). A dynamic risk measure on $\mathcal{T}$ is a sequence of conditional risk measures $\{\rho_{t,T}\}_{t \in \mathcal{T}}$, where for each $t < s$, $t \in \mathcal{T}, s \in \overline{\mathcal{T}}$, the conditional risk measure $\rho_{t,s}$ is a mapping $\rho_{t,s} : L_{t+1,s}^{\infty} \to L_t^{\infty}$.

Thus, $\rho_{t,s}$ associates each stochastic process in $L_{t+1,s}^{\infty}$ with a $\mathcal{F}_t$-measurable random variable. Whenever we write $\rho_{t,s}$, we implicitly assume that $t < s$ with $t \in \mathcal{T}, s \in \overline{\mathcal{T}}$.

Properties 3.1. *A conditional risk measure $\rho_{t,s}$ may satisfy the following properties:*

1. ***Normalisation:*** $\rho_{t,s}(0) := \rho_{t,s}(0, \dots, 0) = 0$.
2. ***Monotonicity:*** $\rho_{t,s}(X_{t+1:s}) \leq \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$ *with* $X_{t+1:s} \leq Y_{t+1:s}$.
3. ***Translation invariance:*** $\rho_{t,s}(X_{t+1:s} + Y_t) = \rho_{t,s}(X_{t+1:s}) + Y_t$, *for all* $X_{t+1:s} \in L_{t+1,s}^{\infty}$ *and* $Y_t \in L_t^{\infty}$.
4. ***Locality:*** $\rho_{t,s}(1_B X_{t+1:s} + 1_{B^{\complement}} Y_{t+1:s}) = 1_B \rho_{t,s}(X_{t+1:s}) + 1_{B^{\complement}} \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$ *and* B *that are* $\mathcal{F}_t$*-measurable.*
5. ***Positive homogeneity:*** $\rho_{t,s}(\lambda X_{t+1:s}) = \lambda \rho_{t,s}(X_{t+1:s})$, *for all* $X_{t+1:s} \in L_{t+1,s}^{\infty}$, *and* $\lambda \in L_t^{\infty}$ *with* $\lambda \geq 0$.
6. ***Convexity:*** $\rho_{t,s}(\lambda X_{t+1:s} + (1-\lambda) Y_{t+1:s}) \leq \lambda \rho_{t,s}(X_{t+1:s}) + (1-\lambda) \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$ *and* $\lambda \in L_t^{\infty}$ *with* $0 \leq \lambda \leq 1$.
7. ***Subadditivity:*** $\rho_{t,s}(X_{t+1:s} + Y_{t+1:s}) \leq \rho_{t,s}(X_{t+1:s}) + \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$.
8. ***Concavity:*** $\rho_{t,s}(\lambda X_{t+1:s} + (1-\lambda) Y_{t+1:s}) \geq \lambda \rho_{t,s}(X_{t+1:s}) + (1-\lambda) \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$ *and* $\lambda \in L_t^{\infty}$ *with* $0 \leq \lambda \leq 1$.
9. ***Superadditivity:*** $\rho_{t,s}(X_{t+1:s} + Y_{t+1:s}) \geq \rho_{t,s}(X_{t+1:s}) + \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$.
10. ***Additivity:*** $\rho_{t,s}(X_{t+1:s} + Y_{t+1:s}) = \rho_{t,s}(X_{t+1:s}) + \rho_{t,s}(Y_{t+1:s})$, *for all* $X_{t+1:s}, Y_{t+1:s} \in L_{t+1,s}^{\infty}$.
11. ***Star-shapedness:*** $\rho_{t,s}(\lambda X_{t+1:s}) \leq \lambda \rho_{t,s}(X_{t+1:s})$, *for all* $X_{t+1:s} \in L_{t+1,s}^{\infty}$ *and* $\lambda \in L_t^{\infty}$ *with* $0 \leq \lambda \leq 1$.

We say a dynamic risk measure $\{\rho_{t,T}\}_{t\in\mathcal{T}}$ satisfies one of the properties if $\rho_{t,T}$ satisfies it for all $t \in \mathcal{T}$. The conditional risk measure $\rho_{t,t+1} : L^\infty_{t+1} \to L^\infty_t$, $t \in \mathcal{T}$, is called a one-step (conditional) risk measure, and we denote it simply by ρ_t, that is, $\rho_t(\cdot) := \rho_{t,t+1}(\cdot)$. The acceptance set of a conditional risk measure $\rho_{t,s}$ is defined by

$$A^\rho_{t,s} := \{X_{t+1:s} \in L^\infty_{t+1,s} \ : \ \rho_{t,s}(X_{t+1:s}) \le 0\}, \tag{9}$$

and we use the notation A^ρ_t for the acceptance set of the one-step risk measure ρ_t.

Next, we define a robustification of strong t.c. dynamic risk measures that are normalised, monotone, and translation invariant. For this we first recall the notion of strong time-consistency and refer to Section 4 for a detailed discussion of notions of time-consistencies.

Definition 3 (Strong Time-Consistency). The dynamic risk measure $\{\rho_{t,T}\}_{t\in\mathcal{T}}$ is strong t.c. if for all $t \in \mathcal{T}$ and $X_{t+1:T} \in L^\infty_{t+1,T}$ it holds that

$$\rho_{t,T}(X_{t+1:T}) = \rho_{t,T}(X_{t+1:s} + \rho_{s,T}(X_{s+1:T})) \quad \forall s \in \{t+1, \dots, T-1\}.$$

It is well known that strong t.c. dynamic risk measures that are normalised, monotone, and translation invariant admit a recursive representation as one-step risk measures, recalled next.

Theorem 1 (Recursive Relation (Cheridito et al. [18], Ruszczyński [48])). *Let $\{\rho_{t,T}\}_{t\in\mathcal{T}}$ be a normalised, monotone, and translation-invariant dynamic risk measure. Then $\{\rho_{t,T}\}_{t\in\mathcal{T}}$ is strong t.c.*[3] *if and only if there exists a family of one-step risk measures $\{\rho_t\}_{t\in\mathcal{T}}$ that are normalised, monotone, and translation invariant, such that for all $t \in \mathcal{T}$ and all $X_{t+1:T} \in L^\infty_{t+1,T}$*

$$\rho_{t,T}(X_{t+1:T}) = \rho_t(X_{t+1} + \rho_{t+1}(X_{t+2} + \cdots + \rho_{T-1}(X_T)\cdots)). \tag{10}$$

As the proposed robustification is defined as the largest (worst-case) value the dynamic risk measure can attain when evaluated at random variables in an uncertainty set, we require the following standing assumption.

Assumption 1. *All considered dynamic uncertainty sets are proper.*

By Theorem 1, a normalised, monotone, translation-invariant, and strong t.c. dynamic risk measure can be represented by a family of one-step risk measures that are normalised, monotone, and translation invariant. Thus, we propose to robustify at each time point the corresponding one-step risk measure by taking the worst case within an uncertainty set.

Definition 4 (Dynamic Robust Risk Measure). Let $\mathfrak{u}$ be a dynamic uncertainty set and $\{\rho_t\}_{t\in\mathcal{T}}$ a family of normalised, monotone, and translation-invariant one-step risk measures. Then we define a **dynamic robust risk measure** $R^{\mathfrak{u},\rho} := \{R^{\mathfrak{u},\rho}_{t,T}\}_{t\in\mathcal{T}}$ on $\mathcal{T}$ by a sequence of conditional robust risk measures $\{R^{\mathfrak{u},\rho}_{t,T}\}_{t\in\mathcal{T}}$, where for each $t<s$, $t \in \mathcal{T}, s \in \overline{\mathcal{T}}$, the conditional robust risk measure $R^{\mathfrak{u},\rho}_{t,s}$ is a mapping $R^{\mathfrak{u},\rho}_{t,s} : L^\infty_{t+1:s} \to L^\infty_t$ given by

$$R^{\mathfrak{u},\rho}_{t,s}(X_{t+1:s}) := \operatorname{ess\,sup}\{\rho_t(Y) \in L^\infty_t : Y \in u_{t+1}(X_{t+1:s})\}.$$

Note that by definition, that is, by taking the essential supremum, each conditional robust risk measure $R^{\mathfrak{u},\rho}_{t,s}$ is $\mathcal{F}_t$-measurable. Moreover, any conditional robust risk measure belongs to the class of conditional risk measures, and we say that a conditional robust risk measure satisfies a property in Properties 3.1, if $R^{\mathfrak{u},\rho}_{t,s}$ satisfies it. Analogous to dynamic risk measures, we call the conditional robust risk measures $R^{\mathfrak{u},\rho}_{t,t+1} : L^\infty_{t+1} \to L^\infty_t$, $t \in \mathcal{T}$, a one-step (conditional) robust risk measure, and we denote it simply by $R^{\mathfrak{u},\rho}_t$, that is, $R^{\mathfrak{u},\rho}_t(\cdot) := R^{\mathfrak{u},\rho}_{t,t+1}(\cdot)$. We drop the superscripts when there is no confusion on the dynamic uncertainty set or the family of one-step risk measures considered. That is, we write $R_{t,s}(\cdot) := R^{\mathfrak{u},\rho}_{t,s}(\cdot)$ and $R := R^{\mathfrak{u},\rho}$.

Lemma 2. *Any dynamic robust risk measure is finite.*

Proof. As $u_{t+1}(X_{t+1:T})$ is bounded, there exist a constant $c \in \mathbb{R}$ such that for all $Y \in u_{t+1}(X_{t+1:T})$, it holds that $Y \le c$. By monotonicity of ρ_t, we obtain that $R^{\mathfrak{u},\rho}_{t,T}(X_{t+1:s}) = \operatorname{ess\,sup}\{\rho_t(Y) \in L^\infty_t : Y \in u_{t+1}(X_{t+1:s})\} \le \rho_t(c) < \infty$. □

A simple example is to set the time-t uncertainty set to be the identity, that is, $u_t(X_{t:T}) = \{X_t\}$, $t \in \overline{\mathcal{T}}$. In this case there is no robustification, as the conditional robust risk measure reduces to the conditional risk measure, that is, $R^{\mathfrak{u},\rho}_t(X_{t+1:T}) = \rho_t(X_{t+1})$.

Next, we provide representations of the acceptance sets of dynamic robust risk measures, illustrating that the dynamic risk measure can be interpreted as a capital requirement.

Proposition 1 (Acceptance Sets). *The acceptance set of a conditional robust risk measure $R^{\mathfrak{u},\rho}_{t,T}$, $t \in \mathcal{T}$, has representation*

$$\begin{aligned} A^R_{t,T} &= \{X_{t+1:T} \in L^\infty_{t+1,T} \ : \ \rho_t(Y) \le 0, \quad \forall Y \in u_{t+1}(X_{t+1:T})\} \\ &= \{X_{t+1:T} \in L^\infty_{t+1,T} \ : \ u_{t+1}(X_{t+1:T}) \subseteq A^\rho_t\}. \end{aligned}$$

If R is translation invariant, it follows that

$$R_{t,T}(X_{t+1:T}) = \operatorname{ess\,inf}\{m \in L^\infty_t \ : \ X_{t+1:T} - m \in A^R_{t,T}\}. \tag{11}$$

Proof. Because $R^{\mathfrak{u},\rho}_{t,T}$ is a conditional robust risk measure, its acceptance set is given by (9). The first representation of the acceptance set follows by the definition of the robust risk measure as an essential supremum over ρ_t, and the second by recalling the acceptance set of one-step risk measures. The last statement follows from translation invariance of R. □

Proposition 1, in particular Equation (11), highlights that R may be interpreted as a robust capital requirement, that is, the minimum amount of capital to be added to a position in order to comply with regulated capital requirements. We observe that the time-t acceptance set of R consists of stochastic processes $X_{t+1:T}$, whose projection at time t, X_t, and any random variable in its uncertainty set is acceptable. Thus, in the case when the dynamic uncertainty set is static, the acceptance set of R is contained in the acceptance set of the one-step risk measure, $A^R_{t,T} \subseteq A^\rho_t$, indicating that fewer positions are acceptable.

When robustifying a normalised, monotone, and translation-invariant dynamic risk measure, some of its properties may get lost. In the next statement we investigate which properties of the dynamic uncertainty sets induce the corresponding properties of the dynamic robust risk measure.

Proposition 2 (Induced Properties). *Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and one-step risk measures $\{\rho_t\}_{t\in\mathcal{T}}$. Then the following holds:*

i. *If $\mathfrak{u}$ is normalised, then R is normalised.*

ii. *If $\mathfrak{u}$ is monotone or order preserving, then R is monotone.*

iii. *If $\mathfrak{u}$ is translation invariant, then R is translation invariant.*

iv. *If $\mathfrak{u}$ is static, then $R_{t,T}(\cdot) = R_t(\cdot)$ for all $t \in \mathcal{T}$.*

v. *If $\mathfrak{u}$ is local, then R is local.*

vi. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be positive homogeneous. If $\mathfrak{u}$ is positive homogeneous, then $R_{t,T}(\lambda X_{t+1:T}) = \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0}R_{t,T}(0)$ for all $0 \le \lambda \in L^\infty_t$ and $X_{t+1:T} \in L^\infty_{t+1,T}$ and $t \in \mathcal{T}$. If, moreover, $\mathfrak{u}$ is normalised, then R is positive homogeneous.*

vii. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be convex. If $u_t(\lambda X_{t:T} + (1-\lambda)Y_{t:T}) \subseteq \lambda u_t(X_{t:T}) + (1-\lambda)u_t(Y_{t:T})$ for all $\lambda \in L^\infty_{t-1}$ with $0 \le \lambda \le 1$ and for all $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$, and $t \in \overline{\mathcal{T}}$, then R is convex.*

viii. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be subadditive. If for all $Z \in u_t(X_{t:T} + Y_{t:T})$ there exists $X' \in u_t(X_{t:T})$ and $Y' \in u_t(Y_{t:T})$ such that $Z \le X' + Y'$, for all $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$ and $t \in \overline{\mathcal{T}}$, then R is subadditive.*

ix. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be subadditive. If $u_t(X_{t:T} + Y_{t:T}) \subseteq u_t(X_{t:T}) + u_t(Y_{t:T})$ for all $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$, and $t \in \overline{\mathcal{T}}$, then R is subadditive.*

x. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be concave. If $\lambda u_t(X_{t:T}) + (1-\lambda)u_t(Y_{t:T}) \subseteq u_t(\lambda X_{t:T} + (1-\lambda)Y_{t:T})$, for all $\lambda \in L^\infty_{t-1}$ with $0 \le \lambda \le 1$ and $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$, and $t \in \overline{\mathcal{T}}$, then R is concave.*

xi. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be superadditive. If for all $Z \in u_t(X_{t:T} + Y_{t:T})$ there exists $X' \in u_t(X_{t:T})$ and $Y' \in u_t(Y_{t:T})$ such that $Z \ge X' + Y'$, for all $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$, and $t \in \overline{\mathcal{T}}$, then R is superadditive.*

xii. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be superadditive. If $u_t(X_{t:T}) + u_t(Y_{t:T}) \subseteq u_t(X_{t:T} + Y_{t:T})$, for all $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$, and $t \in \overline{\mathcal{T}}$, then R is superadditive.*

xiii. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be additive. If $u_t(X_{t:T} + Y_{t:T}) = u_t(X_{t:T}) + u_t(Y_{t:T})$ for all $X_{t:T}$, $Y_{t:T} \in L^\infty_{t,T}$, and $t \in \overline{\mathcal{T}}$, then R is additive.*

xiv. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ and $\mathfrak{u}$ be star-shaped. Then $R_{t,T}(\lambda X_{t+1:T}) \le \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0}R_{t,T}(0)$ for all $\lambda \in L^\infty_t$ with $0 \le \lambda \le 1$, $X_{t:T} \in L^\infty_{t,T}$, and $t \in \mathcal{T}$. If moreover $0 \in u_t(0)$ for all $t \in \overline{\mathcal{T}}$, then R is star-shaped.*

Proof. Throughout the proofs we simply write $R^{\mathfrak{u},\rho}_{t,s}(X_{t+1:s}) = \operatorname{ess\,sup}\{\rho_t(Y) \ : \ Y \in u_{t+1}(X_{t+1:s})\}$, where $\rho_t(Y) \in L^\infty_t$ is understood. Recall that by definition of R the one-step conditional risk measures $\{\rho_t\}_{t\in\mathcal{T}}$ are normalised, monotone, and translation invariant, and thus also local; see, for example, proposition 3.3 in Cheridito et al. [18].

Item (i), because $\mathfrak{u}$ is normalized, it holds that $u_t(0) = 0$ for all $t \in \overline{\mathcal{T}}$. Therefore,

$$R_{t,T}(0) = \operatorname{ess\,sup}\{\rho_t(Y) : \ Y = 0\} = \rho_t(0) = 0, \quad \forall t \in \mathcal{T},$$

and R is normalised.

Item (ii), let $t \in \mathcal{T}$ and $X_{t+1:T} \leq Y_{t+1:T}$, with $X_{t+1:T}, Y_{t+1:T} \in L^\infty_{t+1,T}$. If $\mathfrak{u}$ is monotone, then $u_{t+1}(X_{t+1:T}) \subseteq u_{t+1}(Y_{t+1:T})$. If $\mathfrak{u}$ is order preserving, then for each $Z \in u_{t+1}(X_{t+1:T})$ there exists a $Z' \in u_{t+1}(Y_{t+1:T})$ with $Z \leq Z'$. Moreover, by monotonicity of ρ_t, we have $\rho_t(Z) \leq \rho(Z')$. Thus, in both cases, we obtain

$$\begin{aligned} R_{t,T}(X_{t+1:T}) &= \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(X_{t+1:T})\} \\ &\leq \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(Y_{t+1:T})\} = R_{t,T}(Y_{t+1:T}), \end{aligned}$$

and $R_{t,T}$ is monotone.

Item (iii), let $\mathfrak{u}$ be translation invariant. Then for $t \in \mathcal{T}$ and $Z \in L^\infty_t$, we have $u_{t+1}(X_{t+1:T} + Z) = u_{t+1}(X_{t+1:T}) + Z$ and, using translation invariance of ρ_t, that

$$\begin{aligned} R_{t,T}(X_{t+1:T} + Z) &= \operatorname{ess\,sup}\{\rho_t(Y) : Y \in u_{t+1}(X_{t+1:T}) + Z\} \\ &= \operatorname{ess\,sup}\{\rho_t(Y + Z) : Y \in u_{t+1}(X_{t+1:T})\} \\ &= R_{t,T}(X_{t+1:T}) + Z, \end{aligned}$$

and $R_{t,T}$ is translation invariant.

Item (iv), because $\mathfrak{u}$ is static, we obtain for all $t \in \mathcal{T}$ that $u_{t+1}(X_{t+1:T}) = u_{t+1}(X_{t+1})$ and

$$R_{t,T}(X_{t+1:T}) = \operatorname{ess\,sup}\{\rho_t(Y) : Y \in u_{t+1}(X_{t+1})\} = R_t(X_{t+1}),$$

and $R_{t,T}$ is static.

Item (v), let $\mathfrak{u}$ be local and for $t \in \mathcal{T}$, let $B \in \mathcal{F}_t$ and $X_{t+1:T}, Y_{t+1:T} \in L^\infty_{t+1,T}$. Then we obtain by locality of $\mathfrak{u}$ in the second equation that

$$\begin{aligned} &R_{t,T}(1_B X_{t+1:T} + 1_{B^\complement} Y_{t+1:T}) \\ &\quad = \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(1_B X_{t+1:T} + 1_{B^\complement} Y_{t+1:T})\} \\ &\quad = \operatorname{ess\,sup}\{\rho_t(Z) : Z \in 1_B\, u_{t+1}(X_{t+1:T}) + 1_{B^\complement}\, u_{t+1}(Y_{t+1:T})\} \\ &\quad = \operatorname{ess\,sup}\{\rho_t(X' + Y') : X' \in 1_B\, u_{t+1}(X_{t+1:T}),\ Y' \in 1_{B^\complement}\, u_{t+1}(Y_{t+1:T})\} \\ &\quad = \operatorname{ess\,sup}\{\rho_t(1_B\, X' + 1_{B^\complement}\, Y') : X' \in u_{t+1}(X_{t+1:T}),\ Y' \in u_{t+1}(Y_{t+1:T})\} \\ &\quad = \operatorname{ess\,sup}\{1_B\, \rho_t(X') + 1_{B^\complement}\, \rho_t(Y') : X' \in u_{t+1}(X_{t+1:T}),\ Y' \in u_{t+1}(Y_{t+1:T})\} \\ &\quad = 1_B\, R_{t,T}(X_{t+1:T}) + 1_{B^\complement}\, R_{t,T}(Y_{t+1:T}), \end{aligned}$$

where the fourth equality follows because $X' \in 1_B u_{t+1}(X_{t+1:T})$ implies that $X'(\omega) = 0$ for $\omega \notin B$, and therefore $\{X' : X' \in 1_B u_{t+1}(X_{t+1:T})\} = \{X' 1_B : X' \in u_{t+1}(X_{t+1:T})\}$. The fifth equality holds because ρ_t is local and we conclude that $R_{t,T}$ is local.

The proofs of Items (vi) to (xiii) are delegated to the appendix. □

Remark 1. In this paper, we view a conditional risk measure $\rho_{t,s}$ as a mapping from $L^\infty_{t+1,s}$ to L^∞_t, $s < t$, although some works in the literature define conditional risk measures on the space $L^\infty_{t,s}$, that is, as $\tilde{\rho}_{t,s} : L^\infty_{t,s} \to L^\infty_t$. These two notations are compatible, as by translation invariance of $\rho_{t,s}$, we can set $\tilde{\rho}_{t,s}(X_{t:s}) := \rho_{t,s}(X_{t+1:s}) + X_t$, which is the common definition in the literature. Similarly for robust conditional risk measures, we can define

$$\tilde{R}_{t,s}(X_{t:s}) := R_{t,s}(X_{t+1:s}) + X_t = \operatorname{ess\,sup}\{\rho_t(Y) \in L^\infty_t : Y \in u_{t+1}(X_{t+1:s}) + X_t\}.$$

Note that $\tilde{R}_{t,s}(X_{t:s})$ assesses the risk of $X_{t:s}$ at time t, and that in our definition we view X_t as known and only account for the uncertainty of $X_{t+1:s}$. Therefore, to simplify notation we work with $R_{t,s} : L^\infty_{t+1,s} \to L^\infty_t$. Notice that $R_{t,s}$ satisfies one of the properties in Properties 3.1 if and only if $\tilde{R}_{t,s}$ does.

The next section is devoted to generalising Proposition 2 to if and only if statements. For this, we require the notion of the largest uncertainty set that gives rise to the same dynamic robust risk measures, the consolidated uncertainty set.

3.2. Consolidated Uncertainty Set

From the definition of dynamic robust risk measures, we observe that different choices of dynamic uncertainty sets may lead to the same conditional robust risk measure. Thus, we next introduce the largest uncertainty set that

yields the same dynamic robust risk measure—termed the (dynamic) consolidated uncertainty set. This consolidated uncertainty set will be essential for proving if and only if statements on properties of dynamic robust risk measures.

Definition 5 (Consolidated Uncertainty Set). Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and one-step risk measures $\{\rho_t\}_{t\in\mathcal{T}}$. Its **consolidated uncertainty set** $\mathfrak{U} := \{U_t\}_{t\in\overline{\mathcal{T}}}$ is a collection of time-t uncertainty sets U_t, defined for all $t \in \overline{\mathcal{T}}$ and $X_{t:T} \in L^\infty_{t,T}$ by

$$U_t(X_{t:T}) := \bigcup \{u'_t(X_{t:T}) \subseteq L^\infty_t : \mathfrak{u}' = \{u'_t\}_{t\in\overline{\mathcal{T}}} \qquad R^{\mathfrak{u}'}_{t-1,T}(X_{t:T}) = R^{\mathfrak{u}}_{t-1,T}(X_{t:T})\}.$$

A consolidated uncertainty set, thus, is the largest uncertainty set such that all induced robust risk measures are a.s. equal. The consolidated uncertainty set has the following properties and representations.

Lemma 3 (Representations of Consolidated Uncertainty Sets). *Let R be a dynamic risk measure with uncertainty set $\mathfrak{u}$ and consolidated uncertainty set $\mathfrak{U}$. Then it holds for all $t \in \mathcal{T}$ and $X_{t+1:T} \in L^\infty_{t+1,T}$ that*

i. $U_{t+1}(X_{t+1:T}) = \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{u}}_{t:T}(X_{t+1:T})\}$.
ii. $R^{\mathfrak{U}}_t(X_{t+1:T}) = R^{\mathfrak{u}}_t(X_{t+1:T})$.
iii. *If* $\mathfrak{u}^* := \{u^*_s\}_{s\in\overline{\mathcal{T}}}$ *is an uncertainty set such that* $R^{\mathfrak{u}^*}_{t,T}(X_{t+1:T}) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T})$, *then* $u^*_{t+1}(X_{t+1:T}) \subseteq U_{t+1}(X_{t+1:T})$.
iv. $U_{t+1}(X_{t+1:T}) = A^\rho_t + R^{\mathfrak{U}}_t(X_{t+1:T})$.

Proof. Throughout, we fix $t \in \mathcal{T}$ and $X_{t+1:T} \in L^\infty_{t+1,T}$.

Item (i), let $Z \in \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{u}}_{t:T}(X_{t+1:T})\}$, which implies that $\rho_t(Z) \le R^{\mathfrak{u}}_{t:T}(X_{t+1:T})$. Next, define the dynamic uncertainty set $\mathfrak{u}^\dagger$ by $u^\dagger_{t+1} := u_{t+1} \cup \{Z\}$ and $u^\dagger_s := u_s$ for $s \in \mathcal{T}/\{t+1\}$. As the robust risk measure is defined through an essential supremum, it holds that $R^{\mathfrak{u}^\dagger}_{t:T}(X_{t+1:T}) = R^{\mathfrak{u}}_{t:T}(X_{t+1:T})$. Therefore, we conclude $Z \in U_{t+1}(X_{t+1:T})$.

Conversely, let $Y \in U_{t+1}(X_{t+1:T})$. This means there exists a dynamic uncertainty set $\mathfrak{u}^\dagger := \{u^\dagger_t\}_{t\in\overline{\mathcal{T}}}$ such that $Y \in u^\dagger_{t+1}(X_{t+1:T})$ and $R^{\mathfrak{u}^\dagger}_{t:T}(X_{t+1:T}) = R^{\mathfrak{u}}_{t:T}(X_{t+1:T})$. By definition of the robust risk measure as the essential supremum over elements in the uncertainty set, we have $\rho_t(Y) \le R^{\mathfrak{u}^\dagger}_{t:T}(X_{t+1:T})$ and thus $Y \in \{Z \in L^\infty_{t+1} : \rho_t(Z) \le R^{\mathfrak{u}}_{t:T}(X_{t+1:T})\}$.

Item (ii), using the representation of U_{t+1} from Item (i), we obtain

$$\begin{aligned} R^{\mathfrak{U}}_{t:T}(X_{t+1:T}) &= \operatorname{ess\,sup}\{\rho_t(Y) : Y \in U_{t+1}(X_{t+1:T})\} \\ &= \operatorname{ess\,sup}\{\rho_t(Y) : \rho_t(Y) \le R^{\mathfrak{u}}_{t:T}(X_{t+1:T})\} = R^{\mathfrak{u}}_{t:T}(X_{t+1:T}). \end{aligned}$$

Item (iii), let $\mathfrak{u}^*$ be a dynamic uncertainty set such that $R^{\mathfrak{u}^*}_{t,T}(X_{t+1:T}) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T})$. For $Y \in u^*_{t+1}(X_{t+1:T})$, it holds by definition of the robust risk measure that $\rho_t(Y) \le R^{\mathfrak{u}^*}_{t,T}(X_{t+1:T}) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T})$. Hence, by Item (i) $Y \in U_{t+1}(X_{t+1:T})$.

Item (iv), recall that ρ_t is translation invariant; thus by Item (i)

$$\begin{aligned} U_{t+1}(X_{t+1:T}) &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{U}}_{t:T}(X_{t+1:T})\} \\ &= \{Y \in L^\infty_{t+1} : \rho_t(Y - R^{\mathfrak{U}}_{t:T}(X_{t+1:T})) \le 0\} \\ &= \{Y + R^{\mathfrak{U}}_{t:T}(X_{t+1:T}) \in L^\infty_{t+1} : \rho_t(Y) \le 0\} \\ &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \le 0\} + R^{\mathfrak{U}}_{t:T}(X_{t+1:T}). \\ &= A^\rho_t + R^{\mathfrak{U}}_{t:T}(X_{t+1:T}), \end{aligned}$$

where in the last equation we used the definition of the acceptance set of ρ_t. □

With the consolidated uncertainty set, we can provide necessary and sufficient characterisation of the properties of dynamic robust risk measures. By Lemma 3 (ii) we have $R^{\mathfrak{u}}(\cdot) = R^{\mathfrak{U}}(\cdot)$; thus, whenever we say that R satisfies a property, we implicitly mean that both $R^{\mathfrak{u}}$ and $R^{\mathfrak{U}}$ satisfy it.

Theorem 2. *Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and one-step risk measures $\{\rho_t\}_{t\in\mathcal{T}}$, and denote by $\mathfrak{U}$ the associated consolidated uncertainty set. Then, the following holds:*

1. *R is normalised if and only if $U_{t+1}(0) = A^\rho_t$, for all $t \in \mathcal{T}$.*
2. *R is monotone if and only if $\mathfrak{U}$ is monotone.*
3. *R is translation invariant if and only if $\mathfrak{U}$ is translation invariant.*
4. *R is a family one-step risk measures, that is, $R_{t,T}(\cdot) = R_t(\cdot)$ for all $t \in \mathcal{T}$, if and only if $\mathfrak{U}$ is static.*
5. *R is local if and only if $\mathfrak{U}$ is local.*
6. *Let $\{\rho_t\}_{t\in\mathcal{T}}$ be positive homogeneous. Then R satisfies $R_{t,T}(\lambda X_{t+1:T}) = \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0} R_{t,T}(0)$, for all $0 \le \lambda \in L^\infty_{t-1}$, $X_{t+1:T} \in L^\infty_{t+1,T}$, and $t \in \mathcal{T}$, if and only if $\mathfrak{U}$ is positive homogeneous.*

7. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be convex. Then* R *is convex if and only if* $U_t(\lambda X_{t:T} + (1-\lambda)Y_{t:T}) \subseteq \lambda U_t(X_{t:T}) + (1-\lambda)U_t(Y_{t:T})$ *for all* $\lambda \in L^\infty_{t-1}$ *with* $0 \le \lambda \le 1$, $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$, *and* $t \in \overline{\mathcal{T}}$.

8. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be subadditive. Then* R *is subadditive if and only if* $U_t(X_{t:T} + Y_{t:T}) \subseteq U_t(X_{t:T}) + U_t(Y_{t:T})$ *for all* $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$, *and* $t \in \overline{\mathcal{T}}$.

9. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be additive. R is concave if and only if* $\lambda U_t(X_{t:T}) + (1-\lambda)U_t(Y_{t:T}) \subseteq U_t(\lambda X_{t:T} + (1-\lambda)Y_{t:T})$ *for all* $\lambda \in L^\infty_{t-1}$ *with* $0 \le \lambda \le 1$, $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$, *and* $t \in \overline{\mathcal{T}}$.

10. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be additive. R is superadditive if and only if* $U_t(X_{t:T}) + U_t(Y_{t:T}) \subseteq U_t(X_{t:T} + Y_{t:T})$ *for all* $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$, *and* $t \in \overline{\mathcal{T}}$.

11. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be additive. R is additive if and only if* $U_t(X_{t:T} + Y_{t:T}) = U_t(X_{t:T}) + U_t(Y_{t:T})$ *for all* $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$, *and* $t \in \overline{\mathcal{T}}$.

12. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be positive homogeneous. R satisfies* $R_{t,T}(\lambda X_{t+1:T}) \le \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0}R_{t,T}(0)$ *for all* $X_{t+1:T} \in L^\infty_{t+1,T}$, $\lambda \in L^\infty_t$ *with* $0 \le \lambda \le 1$ *and* $t \in \mathcal{T}$ *if and only if* $\mathfrak{U}$ *is star-shaped. If* $\mathfrak{u}$ *is additionally normalised, then R is star-shaped if and only if* $\mathfrak{U}$ *is star-shaped.*

Proof. The "if" direction of Items 2–12 follows from Proposition 2 by taking the underlying uncertainty set to be $\mathfrak{U}$. For Item 12, note that if ρ_t is positive homogeneous, then it is also star-shaped. For the "if" direction of Item 1, note that if $\mathfrak{U}_{t+1}(X_{t+1:T}) = A^\rho_t$ for all $X_{t+1:T} \in L^\infty_{t+1}$, then by Lemma 3 (iv) $R^{\mathfrak{U}}_{t:T}(0) = 0$ and R is normalised.

Next, we proof the "only if" direction. For this, we fix $t \in \mathcal{T}$ and $X_{0:T} \in L^\infty_{0:T}$.

Item 1, let R be normalised. Then by Lemma 3 (iv), we have

$$U_{t+1}(0) = A^\rho_t + R^{\mathfrak{U}}_t(0) = A^\rho_t.$$

Item 2, let R be monotone, $Y_{t+1:T} \in L^\infty_{t+1,T}$ with $X_{t+1:T} \le Y_{t+1:T}$, and $Z \in U_{t+1}(X_{t+1:T})$. Then, $\rho_t(Z) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) \le R^{\mathfrak{U}}_{t,T}(Y_{t+1:T})$, where the first inequality is due to $Z \in U_{t+1}(X_{t+1:T})$ and the second by monotonicity of R. Therefore $Z \in U_{t+1}(Y_{t+1:T})$ and thus U_{t+1} is monotone.

Item 3, let R be translation invariant and $Z \in L^\infty_t$. Then it holds by Lemma 3 (i) that

$$\begin{aligned}
U_{t+1}(X_{t+1:T} + Z) &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T} + Z)\} \\
&= \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) + Z\} \\
&= \{Y \in L^\infty_{t+1} : \rho_t(Y - Z) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T})\} \\
&= \{Y + Z \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{U}}_{t,T}(X_{t+1:T})\} \\
&= U_{t+1}(X_{t:T}) + Z,
\end{aligned}$$

where the second and third equalities are by translation invariance of R and ρ_t, respectively, and the last equalities are from Lemma 3 (i). Thus, U_{t+1} is translation invariant.

Item 4, let $R_{t,T}(\cdot) = R_t(\cdot)$ for all $t \in \mathcal{T}$; then, $R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) = R^{\mathfrak{U}}_{t,T}(X_{t+1})$ and

$$U_{t+1}(X_{t+1:T}) = \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R^{\mathfrak{U}}_t(X_{t+1})\} = U_{t+1}(X_{t+1}),$$

and U_{t+1} is static.

Item 5, let R be local, $B \in \mathcal{F}_t$, and $Y_{t+1:T} \in L^\infty_{t+1,T}$. Then, using Lemma 3 (iv) in the first and last equation and locality of R in the second, we have that

$$\begin{aligned}
U_{t+1}(1_B X_{t+1:T} + 1_{B^\complement} Y_{t+1:T}) &= A^\rho_t + R^{\mathfrak{U}}_{t,T}(1_B X_{t+1:T} + 1_{B^\complement} Y_{t+1:T}) \\
&= A^\rho_t + 1_B R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) + 1_{B^\complement} R^{\mathfrak{U}}_{t,T}(Y_{t+1:T}) \\
&= 1_B (A^\rho_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T})) + 1_{B^\complement} (A^\rho_t + R^{\mathfrak{U}}_{t,T}(Y_{t+1:T})) \\
&= 1_B U_{t+1}(X_{t+1:T}) + 1_{B^\complement} U_{t+1}(Y_{t+1:T}),
\end{aligned}$$

and U_{t+1} is local.

The proofs of Items 6–12 are delegated to the appendix. □

Although Theorem 2 characterises the properties of R via its consolidated uncertainty set $\mathfrak{U}$, a dynamic robust risk measure is typically defined through a dynamic uncertainty set (e.g., the ones considered in Section 2.3), and not its consolidated one. Thus, we next collect how properties of $\mathfrak{u}$ translate to properties of $\mathfrak{U}$.

Corollary 1. *Let* $\mathfrak{u}$ *be a dynamic uncertainty set with consolidated uncertainty set* $\mathfrak{U}$. *Then,*

1. *If* $\mathfrak{u}$ *satisfies one of the Properties* 2.1 *(i) and (iii)–(viii), then* $\mathfrak{U}$ *satisfies it.*
2. *If* $\mathfrak{u}$ *is normalised, then* $U_{t+1}(0) = A_t^\rho$ *for all* $t \in \mathcal{T}$.
3. *If* $\mathfrak{u}$ *respects order preservation or monotonicity, then* $\mathfrak{U}$ *is monotone and order preserving.*
4. *Let the* $\{\rho_t\}_{t\in\mathcal{T}}$ *be subadditive. If* $Z \in u_t(X_{t:T} + Y_{t:T})$ *implies that there is* $X' \in u_t(X_{t:T})$ *and* $Y' \in u_t(Y_{t:T})$ *such that* $Z \le X' + Y'$, *then* $U_t(X_{t:T} + Y_{t:T}) \subseteq U_t(X_{t:T}) + U_t(Y_{t:T})$.
5. *Let* $\{\rho_t\}_{t\in\mathcal{T}}$ *be additive. If* $Z \ge X' + Y'$ *for any* $X' \in u_t(X_{t:T})$ *and* $Y' \in u_t(Y_{t:T})$ *implies that* $Z \in u_t(X_{t:T} + Y_{t:T})$, *then* $U_t(X_{t:T} + Y_{t:T}) \supseteq U_t(X_{t:T}) + U_t(Y_{t:T})$.

Proof. Properties 2.1 (i), let $\mathfrak{u}$ be proper; because $\mathfrak{u} \subseteq \mathfrak{U}$, we have that $\mathfrak{U}$ is nonempty. Moreover, $\mathfrak{u}$ is bounded from above, which implies that R is also bounded, and thus $\mathfrak{U}$ needs to be bounded from above. For Item 3, note that monotonicity implies the property of order preservation. The other statements follow from the fact that by Proposition 2, a property of $\mathfrak{u}$ implies the corresponding property in R, which by Theorem 2 implies the corresponding property in $\mathfrak{U}$. □

From Corollary 1, we observe that if a dynamic uncertainty set is order preserving, then its consolidated uncertainty set is monotone. Because U_{t+1} contains all $\mathcal{F}_t$-measurable random variables Y with smaller risk (Lemma 3, Item (i)), monotonicity is indeed a desirable property of $\mathfrak{U}$, whereas order preservation is suitable for $\mathfrak{u}$.

Unfortunately, for a well-behaved dynamic uncertainty set $\mathfrak{u}$, one can construct a dynamic uncertainty set $\mathfrak{u}^*$ such that $\mathfrak{U} = \mathfrak{U}^*$, but $\mathfrak{u}^*$ does not satisfy the same properties as $\mathfrak{u}$.

Proposition 3. *Let* $\mathfrak{u}$ *be an uncertainty set and* $\mathfrak{U}$ *the associated consolidated uncertainty set. Then there exists an uncertainty set* $\mathfrak{u}^*$ *such that* $\mathfrak{U} = \mathfrak{U}^*$ *and*

i. $\mathfrak{u}$ *is normalised and* $\mathfrak{u}^*$ *is not normalised.*
ii. $\mathfrak{u}$ *is order preserving and* $\mathfrak{u}^*$ *is not order preserving.*
iii. $\mathfrak{u}$ *is translation invariant and* $\mathfrak{u}^*$ *is not translation invariant.*

Proof. Item (i), take $\mathfrak{u}^* = \mathfrak{U}$.

Item (ii), let $X_{t+1:T} \le Y_{t+1:T}$ and $W \in L^\infty_{t+1}$ be such that $\rho_t(W) = R_{t:T}(Y_{t+1:T})$. Define $u^*_{t+1}(Y_{t+1:T}) := \{W\}$ and $u^*_{t+1}(X_{t+1:T}) := \{Z \in L^\infty_{t+1} : \rho(Z) = R_{t:T}(X_{t+1:T}) \text{ and } Z \not\le W\}$ and $u^*_s := u_s$ for all $s \ne t+1$.

Item (iii), fix $Y_{t:T} \in L^\infty_{t,T}$ and define for all $t \in \overline{\mathcal{T}}$

$$u^*_t(X_{t:T}) := \begin{cases} u_t(X_{t:T}) & X_{t:T} \le Y_{t:T}, \\ U_t(X_{t:T}) & \text{otherwise.} \end{cases}$$

Clearly $\mathfrak{u}^*$ is not translation invariant. □

4. Time-Consistent Dynamic Robust Risk Measures

In the dynamic setting, notions of time-consistency are of utmost importance when, for example, optimising dynamic risk measures; see, for example, Acciaio and Penner [2] and Bielecki et al. [9] for a review. The first key result in this section is Theorem 3, which provides necessary and sufficient criteria for different notions of time-consistencies. The second is Theorem 5, which states that a dynamic robust risk measure is strong t.c. or weak recursive if and only if it can be constructed recursively via a static uncertainty set.

4.1. Notions of Time-Consistencies

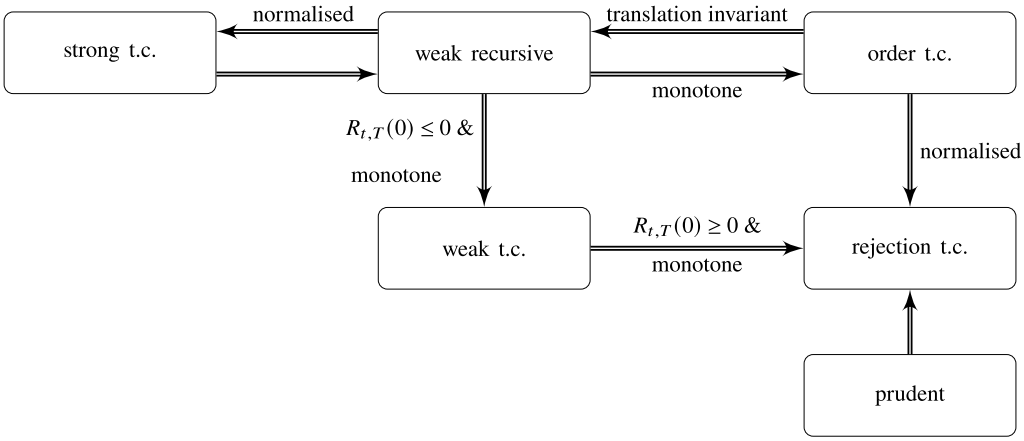
This section is devoted to when time-consistency is preserved by robustification. For this, we first define notions of time-consistencies for dynamic uncertainty sets which we then relate to time-consistencies of dynamic robust risk measures. Several researchers proposed different definitions of time-consistency, and it is not our intention to provide an exhaustive review of this concept, as we put a focus on time-consistencies that result in a recursive representation. We provide interpretation of time-consistencies discussed in this section after their definition and refer the reader to Bielecki et al. [9], who offer an encompassing survey.

Definition 6 (Time-Consistency of Dynamic Uncertainty Sets). Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and one-step risk measures $\{\rho_t\}_{t\in\mathcal{T}}$. Then, $\mathfrak{u}$ is

i. **Strong t.c.**, if for all $t \in \overline{\mathcal{T}}$ and $X_{t:T} \in L^\infty_{t,T}$ it holds that

$$u_t(X_{t:T}) = u_t(X_{t:s} + R_{s,T}(X_{s+1:T})), \quad \forall s \in \{t, \dots, T-1\}.$$

Figure 1. Time-consistencies for a consolidated uncertainty set $\mathfrak{U}$ and for a dynamic robust risk measure R. By Theorem 3, the diagram holds for both $\mathfrak{U}$ and R, with the exception of prudence, which is only defined for $\mathfrak{U}$.



ii. **Order t.c.**, if for all $t \in \overline{\mathcal{T}}$ and $X_{t:T}, Y_{t:T} \in L^{\infty}_{t,T}$ that satisfy

$$X_{t:s} = Y_{t:s} \quad \text{and} \quad u_{s+1}(X_{s+1:T}) \subseteq u_{s+1}(Y_{s+1:T})$$

for some $s \in \{t, \dots, T-1\}$, it holds that

$$u_t(X_{t:T}) \subseteq u_t(Y_{t:T}).$$

iii. **Rejection t.c.**, if for all $t \in \overline{\mathcal{T}}$ and $X_{t:T} \in L^{\infty}_{t,T}$ with $X_t \ge 0$,

$$0 \in u_{t+1}(X_{t+1:T}) \quad \text{implies} \quad 0 \in u_t(X_{t:T}).$$

iv. **Weak recursive**, if for all $t \in \overline{\mathcal{T}}$ and $X_{t:T} \in L^{\infty}_{t,T}$ it holds that

$$u_t(X_{t:T}) = u_t(X_{t:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)), \quad \forall s \in \{t, \dots, T-1\}.$$

v. **Weak t.c.**, if for all $t \in \overline{\mathcal{T}}$ and $X_{t:T} \in L^{\infty}_{t,T}$ it holds that

$$u_t(X_{t:s} + R_{s,T}(X_{s+1:T})) \subseteq u_t(X_{t:T}), \quad \forall s \in \{t, \dots, T-1\}.$$

vi. **Prudent**, if for all $t \in \overline{\mathcal{T}}/\{T\}$ and $X_{t:T} \in L^{\infty}_{t,T}$ it holds that

$$X_t + R_{t,T}(X_{t+1:T}) - R_{t,T}(0) \in u_t(X_{t:T}),$$

and $X_T \in u_T(X_T)$.

Note that prudence implies $X_t \in u_t(X_t)$. Furthermore, in Proposition 4, we show that prudence results in the dynamic robust risk measure always dominating the dynamic risk measure.

Next, we recall different notions of time-consistencies of dynamic risk measures and define a new version called weak recursiveness. We also refer to Figure 1 on how these different time-consistencies are connected.

Definition 7 (Time-Consistency of Dynamic Risk Measures). Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and one-step risk measures $\{\rho_t\}_{t \in \mathcal{T}}$. Then, R is

i. **Order t.c.**, if for all $t \in \mathcal{T}$ and $X_{t+1:T}, Y_{t+1:T} \in L^{\infty}_{t+1,T}$ that satisfy

$$X_{t+1:s} = Y_{t+1:s} \quad \text{and} \quad R_{s,T}(X_{s+1:T}) \le R_{s,T}(Y_{s+1:T})$$

for some $s \in \{t+1, \dots, T-1\}$, it holds that

$$R_{t,T}(X_{t+1:T}) \le R_{t,T}(Y_{t+1:T}).$$

ii. **Rejection t.c.**, if for all $t \in \mathcal{T}$ and $X_{t+1:T} \in L^{\infty}_{t+1,T}$ with $X_{t+1} \ge 0$

$$R_{t+1,T}(X_{t+2:T}) \ge 0 \quad \text{implies} \quad R_{t,T}(X_{t+1:T}) \ge 0.$$

iii. **Weak recursive**, if for all $t \in \mathcal{T}$ and $X_{t+1:T} \in L^{\infty}_{t+1,T}$ it holds that

$$R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)) \quad \forall s \in \{t+1, \dots, T-1\}.$$

iv. **Weak t.c.**, if for all $t \in \mathcal{T}$ and $X_{t+1:T} \in L^{\infty}_{t+1,T}$ it holds that

$$R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T})) \leq R_{t,T}(X_{t+1:T}).$$

Although the literature on time-consistency is extensive, different names and definitions are used and "time-consistency" is often referred to without further distinction, for example, Bion-Nadal and Di Nunno [12] and Cheridito et al. [18]. In many works, the concept of strong t.c. refers to a property where the risk measure exhibits a recursive nature, also called "recursivity" (Bielecki et al. [9], Detlefsen and Scandolo [21], Epstein and Schneider [24]). Order t.c. denotes congruent preferences of decisions with respect to ordering or ranking across different time periods. In the context of preference relations, order t.c. appears in the works of Epstein and Schneider [23], Maccheroni et al. [39], and Wang [54]. Rejection t.c. means that an unacceptable outcome tomorrow is also unacceptable today. In Tutsch [53], this property is referred to as "rejection t.c. with respect to 0," which under translation invariance is equivalent to weak rejection t.c. considered in Bielecki et al. [9]. Although the definition of weak recursiveness is new, a related concept is considered in Bion-Nadal and Di Nunno [12] and Di Nunno and Rosazza Gianin [22], who study fully dynamic risk measures. The property of prudence means that dynamic uncertainty sets contain the future risk they assess.

In the literature, conditional risk measures are often assumed to be normalised, monotone, and translation invariant, in which case strong t.c., order t.c., and weak recursive are equivalent and often referred to simply as "time-consistent" (Delbaen [20], Detlefsen and Scandolo [21], Frittelli and Scandolo [26], Riedel [46], Roorda et al. [47]). However, when robustifying dynamic risk measures these properties may get lost, making it necessary to study them separately. We next relate the new notion of weak recursive to strong and order t.c. For this we make the following observation.

Lemma 4. *Let R be weak recursive and define the normalised version of R by $\tilde{R}_{t,T}(\cdot) := R_{t,T}(\cdot) - R_{t,T}(0)$ for $t \in \mathcal{T}$. Then $\tilde{R}$ is strong t.c.*

As discussed in Section 2.3, many examples of uncertainty sets yield dynamic robust risk measures that are not normalised, for example, induced by the Wasserstein distance (Gao and Kleywegt [28], Luo and Mehrotra [37], Luo and Mehrotra [38], Mohajerin Esfahani and Kuhn [40]). In the context of fully dynamic risk measures, the work of Di Nunno and Rosazza Gianin [22] analyses how concepts of time-consistency, h-longevity, and restriction are connected to normalisation. We provide an example of a dynamic robust risk measure that is strong t.c. but not normalised, in Example 7.

The following alternative characterisation of weak recursiveness illustrates its difference from order time-consistency. Indeed, as seen in Figure 1, neither of these notions implies the other.

Lemma 5. *R is weak recursive if and only if for all $t \in \mathcal{T}$ and $X_{t+1:T}, Y_{t+1:T} \in L^{\infty}_{t+1,T}$ that satisfy*

$$X_{t+1:s} + R_{s,T}(X_{s+1:T}) = Y_{t+1:s} + R_{s,T}(Y_{s+1:T})$$

for all $s \in \{t+1, \dots, T-1\}$, it holds that

$$R_{t,T}(X_{t+1:T}) = R_{t,T}(Y_{t+1:T}).$$

Proof. For the "if" direction let $X_{t+1:T} \in L^{\infty}_{t+1,T}$ and define $Y_{t+1:T} := X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)$, which satisfies, because $Y_{s+1:T} = 0$, that $R_{s,T}(Y_{s+1:T}) = R_{s,T}(0)$. Therefore, $X_{t+1:T}, Y_{t+1:T}$ satisfy that $Y_{t+1:s} + R_{s,T}(Y_{s+1:T}) = X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0) + R_{s,T}(0) = X_{t+1:s} + R_{s,T}(X_{s+1:T})$, and thus the assumption in the statement. Therefore, $R_{t,T}(X_{t+1:T}) = R_{t,T}(Y_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0))$ and R is weak recursive.

For the "only if" direction assume that R is weak recursive and $X_{t+1:T}, Y_{t+1:T} \in L^{\infty}_{t+1,T}$ satisfy $X_{t+1:s} + R_{s,T}(X_{s+1:T}) = Y_{t+1:s} + R_{s,T}(Y_{s+1:T})$. Then we have, for any $s \in \{t+1, \dots, T-1\}$, that

$$\begin{aligned} R_{t,T}(X_{t+1:T}) &= R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)) \\ &= R_{t,T}(Y_{t+1:s} + R_{s,T}(Y_{s+1:T}) - R_{s,T}(0)) = R_{t,T}(Y_{t+1:T}), \end{aligned}$$

which concludes the proof. □

With these notions of time-consistencies at hand, we now state the properties of $\mathfrak{u}$ that yield t.c. dynamic robust risk measures.

Proposition 4 (Induced Time-Consistencies). *Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$. Then, the following holds:*

1. *If $\mathfrak{u}$ is strong t.c., then R is strong t.c.*

2. *Let* $\mathfrak{u}$ *be order t.c. Let* $X_{t+1:T}, Y_{t+1,T} \in L^\infty_{t+1,T}$ *satisfy* $X_{t+1:s} = Y_{t+1:s}$ *and* $u_{s+1}(X_{s+1:T}) \subseteq u_{s+1}(Y_{s+1:T})$*; then it holds that* $R_{t,T}(X_{t+1:T}) \leq R_{t,T}(Y_{t+1:T})$.
3. *If* $\mathfrak{u}$ *is weak recursive, then* R *is weak recursive.*
4. *If* $\mathfrak{u}$ *is weak t.c., then* R *is weak t.c.*
5. *If* $\mathfrak{u}$ *is prudent, then* $R_{t,T}(X_{t+1:T}) \geq \rho_t(X_{t+1} + R_{t+1,T}(X_{t+2:T}))$ *for all* $X_{t+1:T} \in L^\infty_{t+1,T}$*. In particular, it holds that*

$$R_{0,T}(X_{1:T}) \geq \rho_0(X_1 + R_{1,T}(X_{2:T})) \geq \cdots \geq \rho_0 \circ \cdots \circ \rho_{T-1}\left(\sum_{s=1}^{T} X_s\right).$$

Proof. Throughout, we let $t \in \mathcal{T}$ and $X_{0:T} \in L^\infty_{0:T}$.

Item 1, let $\mathfrak{u}$ be strong t.c., then we have for all $s \in \{t+1, \ldots, T-1\}$ that $u_{t+1}(X_{t+1:T}) = u_{t+1}(X_{t+1:s} + R_{s,T}(X_{s+1:T}))$. Because the two uncertainty sets are the same, they give rise to the same robust risk measures, that is, $R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}))$, and R is strong t.c.

Item 2, let $\mathfrak{u}$ be order time-consistency and $X_{t+1:T}, Y_{t+1:T}$ such that $u_{t+1}(X_{t+1:T}) \subseteq u_{t+1}(Y_{t+1:T})$. Then, by definition of R, it holds that $R_{t,T}(X_{t+1:T}) \leq R_{t,T}(Y_{t+1:T})$.

Item 3, let $\mathfrak{u}$ be weak recursive, then $u_{t+1}(X_{t+1:T}) = u_{t+1}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0))$, which implies that the corresponding robust risk measures are equal: that is, that $R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0))$, and R is order t.c.

Item 4, let $\mathfrak{u}$ be weak t.c., then $u_{t+1}(X_{t+1:s} + R_{s,T}(X_{s+1:T})) \subseteq u_{t+1}(X_{t+1:T})$ implies the ordering of the robust risk measures; that is, $R_{t,T}(X_{t+1:T}) \geq R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}))$, and R is weak t.c.

Item 5, let $\mathfrak{u}$ be prudent, then $X_{t+1} + R_{t+1:T}(X_{t+2:T}) \in u_{t+1}(X_{t+1:T})$ and

$$R_{t:T}(X_{t+1:T}) = \operatorname{ess\,sup}\{\rho_t(Y) : Y \in u_{t+1}(X_{t+1:T})\} \geq \rho_t(X_{t+1} + R_{t+1:T}(X_{t+2:T})).$$

Applying the above inequality recursively concludes the proof. □

Next, we provide one of the key results, uniquely connecting time-consistencies of consolidated uncertainty sets with those of the dynamic robust risk measure.

Theorem 3. *Let* R *be a dynamic robust risk measure with dynamic uncertainty set* $\mathfrak{u}$ *and consolidated uncertainty set* $\mathfrak{U}$*. Then, the following holds:*

1. R *is strong t.c. if and only if* $\mathfrak{U}$ *is strong t.c.*
2. R *is order t.c. if and only if* $\mathfrak{U}$ *is order t.c.*
3. R *is rejection t.c. if and only if* $\mathfrak{U}$ *is rejection t.c.*
4. R *is weak recursive if and only if* $\mathfrak{U}$ *is weak recursive.*
5. R *is weak t.c. if and only if* $\mathfrak{U}$ *is weak t.c.*
6. R *satisfies* $R_{t,T}(X_{t+1:T}) \geq \rho_t(X_{t+1} + R_{t+1,T}(X_{t+2:T}))$*, for all* $t \in \mathcal{T}$ *and* $X_{t+1:T} \in L^\infty_{t+1,T}$*, if and only if* $\mathfrak{U}$ *is prudent.*

Proof. The "if" direction of Items 1, 4, 5, and 6 follows from Proposition 4 with $\mathfrak{U} = \mathfrak{u}$. Throughout, we let $t \in \mathcal{T}$ and $X_{0:T} \in L^\infty_{0,T}$.

Item 1, let R be strong t.c., then

$$\begin{aligned} U_{t+1}(X_{t+1:T}) &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:T})\} \\ &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}))\} \\ &= U_{t+1}(X_{t+1:s} + R_{s,T}(X_{s+1:T})), \end{aligned}$$

and $\mathfrak{U}$ is strong t.c.

Item 2, first let $\mathfrak{U}$ be order t.c. and $X_{t+1:s} = Y_{t+1:s}$ with $R^{\mathfrak{U}}_{s,T}(X_{s+1:T}) \leq R^{\mathfrak{U}}_{s,T}(Y_{s+1:T})$. Then $U_{s+1}(X_{s+1:T}) = \{Z \in L^\infty_{s+1} : \rho_s(Z) \leq R^{\mathfrak{U}}_{s,T}(X_{s+1:T})\} \subseteq \{Z \in L^\infty_{s+1} : \rho_s(Z) \leq R^{\mathfrak{U}}_{s,T}(Y_{s+1:T})\} = U_{s+1}(Y_{s+1:T})$. Hence, by order time-consistency of $\mathfrak{U}$, we have $U_{t+1}(X_{t+1:T}) \subseteq U_{t+1}(Y_{t+1:T})$, which implies that $R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) \leq R^{\mathfrak{U}}_{t,T}(Y_{t+1:T})$.

Second, let R be order t.c. and for $s \in \{t+1, \ldots, T-1\}$, let $X_{t+1:s} = Y_{t+1:s}$ with $U_{s+1}(X_{s+1:T}) \subseteq U_{s+1}(Y_{s+1:T})$. Because R is defined through a supremum, we have $R_{s,T}(X_{s+1:T}) \leq R_{s,T}(Y_{s+1:T})$ and, by order time-consistency of R, $R_{t,T}(X_{t+1:T}) \leq R_{t,T}(Y_{t+1:T})$. Hence,

$$\begin{aligned} U_{t+1}(X_{t+1:T}) &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:T})\} \\ &\subseteq \{Y \in L^\infty_{t+1} : \rho_t(Y) \leq R_{t,T}(Y_{t+1:T})\} = U_{t+1}(Y_{t+1:T}), \end{aligned}$$

and $\mathfrak{U}$ is order t.c.

Item 3, note that for any $t \in \mathcal{T}$, we have $0 \in U_{t+1}(X_{t+1:T})$ if and only if $R_{t,T}(X_{t+1:T}) \geq 0$. First, let $\mathfrak{U}$ be rejection t.c. and let $X_{t+1:T} \in L^{\infty}_{t+1,T}$ with $X_{t+1} \geq 0$ and $R_{t+1,T}(X_{t+2:T}) \geq 0$. Then, $0 \in U_{t+2}(X_{t+2:T})$ and by rejection time-consistency of $\mathfrak{U}$, $0 \in U_{t+1}(X_{t+1:T})$. Hence, $R_{t,T}(X_{t+1:T}) \geq 0$ and R is rejection t.c.

Second, let R be rejection t.c. and let $X_{t:T} \in L^{\infty}_{t,T}$ with $X_t \geq 0$ and $0 \in U_{t+1}(X_{t+1:T})$. Then, $R_{t,T}(X_{t+1:T}) \geq 0$ and by rejection time-consistency of R, $R_{t-1,T}(X_{t:T}) \geq 0$. Hence, $0 \in U_t(X_{t:T})$ and $\mathfrak{U}$ is rejection t.c.

Item 4, let R be weak recursive. Therefore, all $s \in \{t+1, \dots, T-1\}$,

$$\begin{aligned} U_{t+1}(X_{t+1:T}) &= \{Y \in L^{\infty}_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:T})\} \\ &= \{Y \in L^{\infty}_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0))\} \\ &= U_{t+1}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)), \end{aligned}$$

and $\mathfrak{U}$ is weak recursive.

Item 5, let R be weak t.c., then for all $s \in \{t+1, \dots, T-1\}$,

$$\begin{aligned} U_{t+1}(X_{t+1:T}) &= \{Y \in L^{\infty}_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:T})\} \\ &\supseteq \{Y \in L^{\infty}_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}))\} \\ &= U_{t+1}(X_{t+1:s} + R_{s,T}(X_{s+1:T})), \end{aligned}$$

and $\mathfrak{U}$ is weak t.c.

Item 6, let R satisfy $R_{t,T}(X_{t+1:T}) \geq \rho_t(X_{t+1} + R_{t+1,T}(X_{t+2:T}))$. Then, we have that $X_{t+1} + R_{t+1,T}(X_{t+2:T}) \in \{Y \in L^{\infty}_{t+1} : \rho_t(Y) \leq R_{t,T}(X_{t+1:T})\} = U_{t+1}(X_{t+1:T})$ and $\mathfrak{U}$ is prudent. □

The next corollary collects which notions of time-consistencies of $\mathfrak{U}$ are implied by those of $\mathfrak{u}$.

Corollary 2. *Let $\mathfrak{u}$ be a dynamic uncertainty set with consolidated uncertainty set $\mathfrak{U}$. If $\mathfrak{u}$ satisfies one of the time-consistencies in Definition 6 (i), (iv), (v), and (vi), then $\mathfrak{U}$ satisfies it.*

Proof. By Proposition 4, each property in $\mathfrak{u}$ implies the analogous property in R, which by Theorem 3 implies the corresponding property in $\mathfrak{U}$. □

We conclude this section by providing connections and implications of the different notions of time-consistencies, which are illustrated in Figure 1. Note that by Theorem 3 most of the statements also hold for dynamic robust risk measures.

Proposition 5 (Time-Consistencies of Consolidated Uncertainty Sets). *Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and consolidated uncertainty set $\mathfrak{U}$. Then it holds that*

1. *If $\mathfrak{U}$ is strong t.c., then it is weak recursive.*
2. *Let $\mathfrak{U}$ satisfy $U_{t+1}(0) = A^{\rho}_t$ for all $t \in \mathcal{T}$. If $\mathfrak{U}$ is weak recursive, then it is strong time-consistent.*
3. *If $\mathfrak{U}$ is strong t.c., then $R_{t,T}(X_{t+1:T} + \lambda R_{s,T}(0)) = R_{t,T}(X_{t+1:T})$, for all $X_{t+1:T} \in L^{\infty}_{t+1,T}$ and $\lambda \in \mathbb{Z}$.*
4. *If $\mathfrak{U}$ is monotone and weak recursive, then it is order t.c.*
5. *If $\mathfrak{U}$ is translation invariant and order t.c., then it is weak recursive.*
6. *Let $R_{t,T}(0) \leq 0$ for all $t \in \mathcal{T}$. If $\mathfrak{U}$ is monotone and weak recursive, then it is weak t.c.*
7. *Let $\mathfrak{U}$ be monotone and $0 \in U_t(0)$ for all $t \in \overline{\mathcal{T}}$. If $\mathfrak{U}$ is weak t.c., then it is rejection t.c.*
8. *If $\mathfrak{U}$ is prudent, then it is rejection t.c.*
9. *Let $U_{t+1}(0) = A^{\rho}_t$ for all $t \in \mathcal{T}$. If $\mathfrak{U}$ is order t.c., then it is rejection t.c.*

Proof. Item 1, let $\mathfrak{U}$ be strong t.c., then, by Theorem 3 Item 1, R is strong t.c. Next, let $X_{t+1:T}, Y_{t+1:T} \in L^{\infty}_{t+1,T}$ satisfy $X_{t+1:s} + R_{s,T}(X_{s+1:T}) = Y_{t+1:s} + R_{s,T}(Y_{s+1:T})$ for all $s \in \{t+1, \dots, T-1\}$. Then, by strong time-consistency of R,

$$R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T})) = R_{t,T}(Y_{t+1:s} + R_{s,T}(Y_{s+1:T})) = R_{t,T}(Y_{t+1:T}).$$

Thus, by Lemma 5, R is weak recursive, and applying Theorem 3 Item 4 yields the claim.

Item 2, by Theorem 2, $U_{t+1}(0) = A^{\rho}_t$ is equivalent to R being normalised. Moreover, if R is normalised, then strong time-consistency and weak recursive are equivalent.

Item 3, by Item 1 R is weak recursive. Next, fix $X_{t+1:T} \in L^{\infty}_{t+1,T}$ and define $Y_{t+1:T} := X_{t+1:T} - R_{s,T}(0)$. Then, we obtain by first applying weak recursiveness, then the definition of Y, and strong t.c.,

$$R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)) \tag{12a}$$

$$= R_{t,T}(Y_{t+1:s} + R_{s,T}(Y_{s+1:T})) \tag{12b}$$

$$= R_{t,T}(Y_{t+1:T}) \tag{12c}$$

$$= R_{t,T}(X_{t+1:T} - R_{s,T}(0)). \tag{12d}$$

Iteratively applying Equations (12) to $\tilde{X}_{t+1:T} := X_{t+1:T} - R_{s,T}(0)$ yields that $R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:T} - \lambda R_{s,T}(0))$, for any positive integer λ.

Next, we apply Equations (12) to $\tilde{X}_{t+1:T} := X_{t+1:T} + R_{s,T}(0)$, which gives

$$R_{t,T}(X_{t+1:T} + R_{s,T}(0)) = R_{t,T}(X_{t+1:T} + R_{s,T}(0) - R_{s,T}(0)) = R_{t,T}(X_{t+1:T}). \tag{13}$$

Iteratively applying Equation (13) to $\tilde{X}_{t+1:T} := X_{t+1:T} + R_{s,T}(0)$ yields that $R_{t,T}(X_{t+1:T}) = R_{t,T}(X_{t+1:T} + \lambda R_{s,T}(0))$, for any positive integer λ. Combining these results concludes the proof.

Item 4, for $t \le s$, let $X_{t:s} = Y_{t:s}$ such that $U_{s+1}(X_{s+1:T}) \subseteq U_{s+1}(Y_{s+1:T})$. This implies that $R_{s,T}(X_{s+1:T}) \le R_{s,T}(Y_{s+1:T})$ and therefore

$$X_{t:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0) \le Y_{t:s} + R_{s,T}(Y_{s+1:T}) - R_{s,T}(0).$$

By monotonicity and weak recursiveness of $\mathfrak{U}$, it holds that

$$\begin{aligned} U_t(X_{t:T}) &= U_t(X_{t:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)) \\ &\subseteq U_t(Y_{t:s} + R_{s,T}(Y_{s+1:T}) - R_{s,T}(0)) = U_t(Y_{t:T}), \end{aligned}$$

which implies that $\mathfrak{U}$ is order t.c.

Item 5, by Theorems 2 and 3, R is translation invariant and order t.c. We show that R is weak recursive. For $s \in \{t+1, \dots T-1\}$ define $Y_{t+1:T}$ by $Y_{t+1:s} := X_{t+1:s}$, $Y_{s+1} := R_{s,T}(X_{s+1:T}) - R_{s,T}(0)$, and $Y_{s+2:T} := 0$. Then, by translation invariance of R and noting that R is not necessarily normalised, we obtain

$$\begin{aligned} R_{s,T}(Y_{s+1:T}) &= R_{s,T}(R_{s,T}(X_{s+1:T}) - R_{s,T}(0)) \\ &= R_{s,T}(0) + R_{s,T}(X_{s+1:T}) - R_{s,T}(0) = R_{s,T}(X_{s+1:T}). \end{aligned}$$

Hence, $Y_{t+1:s} = X_{t+1:s}$ and $R_{s,T}(X_{s+1:T}) = R_{s,T}(Y_{s+1:T})$; thus, by order time-consistency of R it holds that

$$R_{t,T}(X_{t+1:T}) = R_{t,T}(Y_{t+1:T}) = R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)),$$

and R is weak recursive. Theorem 3 concludes the proof.

Item 6, by weak recursive of $\mathfrak{U}$, we have for all $s \in \{t, \dots, T-1\}$ that $U_t(X_{t:T}) = U_t(X_{t:s} + R_{s,T}(X_{s+1:T}) - R_{s,T}(0)) \supseteq U_t(X_{t:s} + R_{s,T}(X_{s+1:T}))$, where the set inclusion is due to monotonicity of $\mathfrak{U}$ and because $R_{s,T}(0) \le 0$. Thus, we conclude that $\mathfrak{U}$ is weak t.c.

Item 7, note that by Theorems 2 and 3, R is monotone and weak t.c. Let $0 \in U_{t+2}(X_{t+2:T})$, and $X_{t+1} \ge 0$. This implies, by normalisation of ρ_{t+1}, that

$$R_{t+1,T}(X_{t+2:T}) = \sup\{\rho_{t+1}(Y) : Y \in U_{t+2}(X_{t+2:T})\} \ge \rho_{t+1}(0) = 0. \tag{14}$$

Next, by weak time-consistency (with $s = t+1$), then monotonicity of R, and then applying (14), we have

$$\begin{aligned} R_{t,T}(X_{t+1:T}) &\ge R_{t,T}(X_{t+1} + R_{t+1,T}(X_{t+2:T})) \\ &\ge R_{t,T}(R_{t+1,T}(X_{t+2:T})) \\ &\ge R_{t,T}(0) \ge 0, \end{aligned}$$

where the last inequality follows because by assumption $0 \in U_s(0)$, for all $s \in \overline{T}$. Thus, we conclude that $\mathfrak{U}$ is rejection t.c.

Item 8, first note that prudence implies $R_t(X_{t+1:T}) \ge \rho_t(X_{t+1} + R_{t+1}(X_{t+2:T}))$. Next, let $R_{t+1,T}(X_{t+2:T}) \ge 0$ and $X_{t+1} \ge 0$; then monotonicity and normalisation of ρ_t imply that $R_t(X_{t+1:T}) \ge \rho_t(X_{t+1} + R_{t+1}(X_{t+2:T})) \ge \rho_t(0) = 0$. We conclude that R and hence $\mathfrak{U}$ are rejection t.c.

Item 9, let $0 \in U_{t+1}(X_{t+1:T})$ and $X_t \ge 0$. Recall that by Theorem 2, $U_{t+1}(0) = A_{\rho_{t+1}}$ implies that R is normalised. Therefore, $R_{t,T}(0) = 0 \le R_{t,T}(X_{t+1:T})$, where the inequality follows because $0 \in U_{t+1}(X_{t+1:T})$. Hence, $U_{t+1}(0) \subseteq U_{t+1}(X_{t+1:T})$. Next, define $X'_{t:T} := (0, X_{t+1:T})$ and $Y_{t:T} := 0$. Then clearly $Y_t = 0 = X_t$ and $U_{t+1}(Y_{t+1:T}) \subseteq U_{t+1}(X_{t+1:T})$; thus, by order time-consistency of $\mathfrak{U}$, we have that $U_t(Y_{t:T}) = U_t(0) \subseteq U_t(X_{t:T})$. By assumption, we have $U_t(0) = A_t^\rho \ni 0$, where zero is an element of A_t^ρ because ρ_t is normalised. We conclude that $\mathfrak{U}$ is rejection t.c. □

4.2. Construction of t.c. Dynamic Robust Risk Measures

In this section we investigate the intrinsic connection between static uncertainty sets and t.c. dynamic uncertainty sets. To facilitate notation, we denote by $\mathfrak{u}^{\varsigma}$ a dynamic uncertainty set that possesses the property of being static. Recall that by Corollary 1 the corresponding consolidated uncertainty set $\mathfrak{U}^{\varsigma}$ is also static.

The first result is negative, in that static uncertainty sets cannot give rise to (most of the notions of) t.c. dynamic robust risk measures. However, one of the key results of this section (Theorem 5) is that dynamic robust risk measures are weak recursive if and only if they admit a recursive representation of one-step robust conditional risk measures that arise from a static uncertainty set, generalising Theorem 1 to the robust setting. To establish this, we require the following result on the connection of static and t.c. dynamic uncertainty sets.

Proposition 6. *Let R be a dynamic robust risk measure with dynamic uncertainty set $\mathfrak{u}$ and consolidated uncertainty set $\mathfrak{U}$. Then the following hold:*

1. *Let $R_{t,T}$ be surjective for all $t \in \mathcal{T}$. If $\mathfrak{u}^{\varsigma}$ is static, then $\mathfrak{U}^{\varsigma}$ and $R^{\mathfrak{U}^{\varsigma}}$ do not satisfy the time-consistency notions of Items (i) (iv), and (v) of Definitions 6 and 7 and Item (vi) of Definition 6.*

2. *$R^{\mathfrak{U}}$ or equivalently $\mathfrak{U}$ is weak recursive if and only if there exists a static consolidated uncertainty set $\mathfrak{U}^{\varsigma} := \{U_t^{\varsigma}\}_{t\in\overline{\mathcal{T}}}$ satisfying recursively backward in time*

$$U_T(X_T) = U_T^{\varsigma}(X_T), \quad \textit{and} \tag{15a}$$

$$U_t(X_{t:T}) = U_t^{\varsigma}(X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{U}}_{t,T}(0)), \quad \forall t \in \overline{\mathcal{T}} \setminus \{T\}. \tag{15b}$$

3. *If $R^{\mathfrak{U}}$ or equivalently $\mathfrak{U}$ is strong t.c., then there exists a static consolidated uncertainty set $\mathfrak{U}^{\varsigma} := \{U_t^{\varsigma}\}_{t\in\overline{\mathcal{T}}}$ satisfying recursively backward in time*

$$U_T(X_T) = U_T^{\varsigma}(X_T), \quad \textit{and}$$

$$U_t(X_{t:T}) = U_t^{\varsigma}(X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T})), \quad \forall t \in \overline{\mathcal{T}} \setminus \{T\}.$$

Proof. Item 1, let $\mathfrak{u}^{\varsigma}$ be static and strong t.c. and fix $X_t \in L_t^{\infty}$. Then it follows that $u_t^{\varsigma}(X_t) = u_t^{\varsigma}(X_{t:T}) = u_t^{\varsigma}(X_t + R_{t,T}(X_{t+1:T}))$ where $X_{t+1:T}$ is an arbitrary process in $L^{\infty}_{t+1,T}$. Moreover, surjectivity of $R_{t,T}$ implies that for any $Y_t \in L_t^{\infty}$ there exists an $X_{t+1:T} \in L^{\infty}_{t+1,T}$ such that $Y_t = X_t + R_{t,T}(X_{t+1:T})$. Thus, $u_t^{\varsigma}(X_t) = u_t^{\varsigma}(Y_t)$ for any $X_t, Y_t \in L_t^{\infty}$. This implies that $R_{t-1,T}(X_{t:T}) = C \in L_{t-1}^{\infty}$ for all $X_{t:T} \in L^{\infty}_{t,T}$, which, by surjectivity of $R_{t,T}$, cannot be true. The same reasoning holds if $\mathfrak{u}$ is weak recursive.

Let $\mathfrak{u}^{\varsigma}$ be static and weak t.c. Then $u_t^{\varsigma}(X_t) = u_t^{\varsigma}(X_{t:T}) \supseteq u_t^{\varsigma}(X_t + R_{t,T}(X_{t+1:T})) = u_t^{\varsigma}(Y_t)$ for all $X_t, Y_t \in L_t^{\infty}$. Reversing the role of X_t and Y_t, it follows that $u_t^{\varsigma}(X_t) = u_t^{\varsigma}(Y_t)$ for all $X_t, Y_t \in L_t^{\infty}$. This implies that R is a constant contradicting surjectivity.

Finally, let $\mathfrak{u}^{\varsigma}$ be static and prudent. By surjectivity of R, for any $Y_t \in L_t^{\infty}$ there exists a $X_{t+1:T} \in L^{\infty}_{t+1,T}$ such that $Y_t = R_{t,T}(X_{t+1:T})$. Then, by prudence of $\mathfrak{u}^{\varsigma}$, we have $X_t + Y_t = X_t + R_{t,T}(X_{t+1:T}) \in u_t^{\varsigma}(X_{t:T}) = u_t^{\varsigma}(X_t)$ for all $X_t, Y_t \in L_t^{\infty}$. Thus, we conclude $u_t^{\varsigma}(X_t) = L_t^{\infty}$ for all $X_t \in L_t^{\infty}$, which implies that $\mathfrak{u}$ cannot be proper.

Item 2, let $\mathfrak{U}$ be weak recursive and define for all $X_{t:T} \in L^{\infty}_{t,T}$ the dynamic uncertainty set $\mathfrak{U}^{\varsigma} := \{U_t^{\varsigma}\}_{t\in\overline{\mathcal{T}}}$ via

$$U_t^{\varsigma}(X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{U}}_{t,T}(0)) := U_t(X_{t:T}) = U_t(X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{U}}_{t,T}(0)),$$

where the second equality follows because $\mathfrak{U}$ is weak recursive. Furthermore, $\mathfrak{U}^{\varsigma}$ is static because for any $Y_t \in L_t^{\infty}$, there exists a $X_{t:T} \in L^{\infty}_{t,T}$ such that $Y_t = X_t + R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0)$. Note that we do not need R to be surjective, as we can choose $X_t \in L_t^{\infty}$ arbitrarily.

Conversely, let $\mathfrak{U}$ be given and let $\mathfrak{U}^{\varsigma}$ be a static uncertainty satisfying (15a) recursively backward in time. Define $Y_{t:T}$ by $Y_t := X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{U}}_{t,T}(0)$ and $Y_{t+1:T} := 0$. Then, applying Equation (15a) first to $Y_{t:T}$ and in the last equation to $X_{t:T}$, we obtain

$$\begin{aligned} U_t(X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{U}}_{t,T}(0)) &= U_t(Y_{t:T}) \\ &= U_t^{\varsigma}(Y_t + R^{\mathfrak{U}}_{t,T}(0) - R^{\mathfrak{U}}_{t,T}(0)) \\ &= U_t^{\varsigma}(X_t + R^{\mathfrak{U}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{U}}_{t,T}(0)) \\ &= U_t(X_{t:T}). \end{aligned}$$

Thus, $\mathfrak{U}$ is weak recursive.

Item 3 follows using similar reasoning as Item 2. □

Proposition 6 states that the consolidated uncertainty set of any strong t.c. or weak recursive dynamic robust risk measure can be represented via static uncertainty sets. This provides a way to construct t.c. dynamic robust risk measures from static uncertainty sets. Moreover, the resulting dynamic risk measure can be seen as recursively applying a robust conditional risk measure arising from a static uncertainty set.

Theorem 4 (Construction of Dynamic Uncertainty Sets). *Let* $\mathfrak{u}^\varsigma$ *be a dynamic uncertainty set that is static and define a dynamic uncertainty set* $\mathfrak{u} := \{u_t\}_{t\in\overline{\mathcal{T}}}$ *recursively backward in time*

$$u_T(X_T) := u_T^\varsigma(X_T), \quad \textit{and}$$

$$u_t(X_{t:T}) := u_t^\varsigma(X_t + R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0)), \quad \forall t \in \overline{\mathcal{T}} \setminus \{T\} \quad \textit{and} \quad X_{t:T} \in L^\infty_{t,T}.$$

Then, it holds that $R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) = R^{\mathfrak{u}^\varsigma}_t(X_{t+1} + R^{\mathfrak{u}}_{t+1,T}(X_{t+2:T}) - R^{\mathfrak{u}}_{t+1,T}(0))$ *and*

$$R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) = R^{\mathfrak{u}^\varsigma}_t(Y_{t+1} + R^{\mathfrak{u}^\varsigma}_{t+1}(Y_{t+2} + R^{\mathfrak{u}^\varsigma}_{t+2}(Y_{t+3} + \ldots + R^{\mathfrak{u}^\varsigma}_{T-1}(Y_T)\ldots))),$$

where $Y_t = X_t - R^{\mathfrak{u}^\varsigma}_t(0)$ *for all* $t \in \mathcal{T}$.

Denote by $\mathfrak{U}$ *the consolidated uncertainty set of* $\mathfrak{u}$, *then*

i. $R^{\mathfrak{u}}$ *and* $\mathfrak{U}$ *are weak recursive.*

ii. *If* $\mathfrak{u}$ *is normalised, then* $R^{\mathfrak{u}}$ *and* $\mathfrak{U}$ *are strong t.c.*

iii. $\mathfrak{u}$ *is normalised if and only if* $\mathfrak{u}^\varsigma$ *is normalised.*

iv. $\mathfrak{u}$ *is translation invariant if and only* $\mathfrak{u}^\varsigma$ *is translation invariant.*

v. $\mathfrak{u}$ *is order preserving if and only if* $\mathfrak{u}^\varsigma$ *is order preserving.*

Proof. Let $\mathfrak{U}^\varsigma$ be the consolidated uncertainty set of $\mathfrak{u}^\varsigma$. Then by Corollary 1, $\mathfrak{U}^\varsigma$ is static and $\mathfrak{U}$ satisfies $U_t(X_{t:T}) = U_t^\varsigma(X_t + R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0))$ for all $t \in \overline{\mathcal{T}} \setminus \{T\}$. The property $R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) = R^{\mathfrak{u}^\varsigma}_t(X_{t+1} - R^{\mathfrak{u}}_{t+1,T}(0) + R^{\mathfrak{u}}_{t+1,T}(X_{t+2:T}))$ follows by definition of the dynamic robust risk measure and noting that $\mathfrak{U}^\varsigma$ is static. The recursion follows by applying this property recursively and because $u_t(0) = u_t^\varsigma(0)$ for all $t \in \overline{\mathcal{T}}$.

Item (i), weak recursive of $R^{\mathfrak{u}}$ and $\mathfrak{U}$ follows from Proposition 6.

Item (ii), if $\mathfrak{u}$ is normalised, then by Corollary 1 $\mathfrak{U}$ satisfies $U_{t+1}(0) = A_t^\rho$ for all $t \in \mathcal{T}$. Applying Proposition 5 gives that $\mathfrak{U}$ and thus also $R^{\mathfrak{u}}$ are strong time-consistency.

Item (iii), if $\mathfrak{u}^\varsigma$ is normalised, then $\mathfrak{u}$ is by definition normalised. Next, assume that $\mathfrak{u}$ is normalised, then

$$0 = u_t(0) = u_t^\varsigma(0 + R^{\mathfrak{u}}_{t,T}(0) - R^{\mathfrak{u}}_{t,T}(0)) = u_t^\varsigma(0),$$

and $\mathfrak{u}^\varsigma$ is normalised.

Item (iv), let $\mathfrak{u}^\varsigma$ be translation invariant, then we have for any $Z \in L^\infty_s$, $s < t$, that

$$\begin{aligned} u_t(X_{t:T}) + Z &= u_t^\varsigma(X_t + R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0)) + Z \\ &= u_t^\varsigma(X_t + R^{\mathfrak{u}}_{t,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0) + Z) = u_t(X_{t:T} + Z), \end{aligned}$$

and $\mathfrak{u}^\varsigma$ is translation invariant. The other direction follows using similar steps.

Item (v), let $\mathfrak{u}^\varsigma$ be order preserving and $X_{t:T} \le Y_{t:T}$. Then, $u_T(Z_T) = u_T^\varsigma(Z_T)$; thus, u_T is order preserving, and, by Proposition 2, $R^{\mathfrak{u}}_{T-1,T}$ is monotone. Thus,

$$X_{T-1} + R^{\mathfrak{u}}_{T-1,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{T-1,T}(0) \le Y_{T-1} + R^{\mathfrak{u}}_{T-1,T}(Y_{t+1:T}) - R^{\mathfrak{u}}_{T-1,T}(0).$$

Next, let $Z \in u_{T-1}(X_{T-1,T}) = u_{T-1}^\varsigma(X_{T-1} + R^{\mathfrak{u}}_{T-1,T}(X_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0))$, then because u_{T-1}^ς is order preserving, there exists a $W \in u_{T-1}^\varsigma(Y_{T-1} + R^{\mathfrak{u}}_{T-1,T}(Y_{t+1:T}) - R^{\mathfrak{u}}_{t,T}(0)) = u_{T-1}(Y_{T-1:T})$ such that $Z \le W$, and thus u_{T-1} is order preserving. Applying the same reasoning recursively backward in time yields that $\mathfrak{u}$.

For the "only if" direction, we show that if $\mathfrak{u}^\varsigma$ is not order preserving, then $\mathfrak{u}$ is also not order preserving. For this, let $\mathfrak{u}^\varsigma$ be not order preserving. That is, for some $t \in \overline{\mathcal{T}}$ and $X_t \le Y_t$, there does not exist $W \in u_t^\varsigma(Y_t)$ satisfying $Z \le W$ for all $Z \in u_t^\varsigma(X_t)$. Now define $X_t' := X_t + R^{\mathfrak{u}}_{t,T}(0)$ and $Y_t' := Y_t + R^{\mathfrak{u}}_{t,T}(0)$, which satisfy $X_t' \le Y_t'$. Then

$$u_t(X_t') = u_t(X_t + R^{\mathfrak{u}}_{t,T}(0)) = u_t^\varsigma(X_t)$$

and similarly $u_t(Y_t') = u_t^\varsigma(Y_t')$. Because $\mathfrak{u}$ is not order preserving, there does not exist a $W \in u_t^\varsigma(Y_t) = u_t(Y_t')$ such that $Z \le W$, for all $Z \in u_t^\varsigma(X_t) = u_t(X_t')$. Hence, u_t is not order preserving. □

We further obtain that any static uncertainty set gives rise to a t.c. dynamic robust risk measure via a recursive representation.

Theorem 5 (Recursive Relation). *Let R be a dynamic robust risk measure. Then the following holds:*

i. *R is weak recursive if and only if there exists a static uncertainty set $\mathfrak{u}^\varsigma := \{u_t^\varsigma\}_{t\in\overline{\mathcal{T}}}$ such that for all $t \in \mathcal{T}$ and $X_{t+1:T} \in L^\infty_{t+1,T}$*

$$R_{t,T}(X_{t+1:T}) = R_t^{\mathfrak{u}^\varsigma}(Y_{t+1} + R_{t+1}^{\mathfrak{u}^\varsigma}(Y_{t+2} + R_{t+2}^{\mathfrak{u}^\varsigma}(Y_{t+3} + \ldots + R_{T-1}^{\mathfrak{u}^\varsigma}(Y_T)\ldots))), \tag{16}$$

where $Y_t := X_t - R_t^{\mathfrak{u}^\varsigma}(0)$, $t \in \mathcal{T}$, and $Y_T := X_T$.

ii. *R is normalised and strong t.c. if and only if there exists a static and normalised uncertainty set $\mathfrak{u}^\varsigma := \{u_t^\varsigma\}_{t\in\overline{\mathcal{T}}}$ such that Equation (16) holds with $Y_t := X_t$, $t \in \overline{\mathcal{T}}$.*

Proof. The "if" direction of both parts is a consequence of Theorem 4. For the "only if" direction of (i), let $R^{\mathfrak{u}}$ be weak recursive; then by Proposition 6 Item 2 there exists a static consolidated uncertainty set $\mathfrak{U}^\varsigma$ that satisfies Equation (15a). Because $\mathfrak{U}^\varsigma$ is proper, we can define $\mathfrak{u}^\varsigma := \mathfrak{U}^\varsigma$, which is static and proper. Applying Theorem 4, we obtain the recursion.

For the "only if" direction of (ii) let $R^{\mathfrak{u}}$ be strong t.c.; then there exists by Proposition 6 Item 3 a static consolidated uncertainty set $\mathfrak{U}^\varsigma := \{U_t^\varsigma\}_{t\in\overline{\mathcal{T}}}$ satisfying

$$U_t(X_{t:T}) = U_t^\varsigma(X_t + R_{t,T}^{\mathfrak{u}}(X_{t+1:T})), \quad \forall t \in \overline{\mathcal{T}} \setminus \{T\}. \tag{17}$$

Because $\mathfrak{U}^\varsigma$ is proper, we can define for all $t \in \mathcal{T}$

$$\mathfrak{u}^\varsigma(X_t) := \mathfrak{U}^\varsigma(X_t), \quad \text{if} \quad X_t \neq 0, \quad \text{and} \quad \mathfrak{u}^\varsigma(0) := 0.$$

Clearly, $\mathfrak{u}^\varsigma$ is normalised and proper and its consolidated uncertainty set is $\mathfrak{U}^\varsigma$, which satisfies (17). Theorem 4 provides the recursion. □

The next results uniquely connect dynamic uncertainty sets that satisfy the recursive relation.

Proposition 7. *Let $\mathfrak{u}$ and $\mathfrak{u}'$ be two dynamic uncertainty sets. If for all $t \in \mathcal{T}$ it holds that*

$$R_{t,T}^{\mathfrak{u}}(X_{t+1:T}) = R_t^{\mathfrak{u}'}(X_{t+1} + R_{t+1}^{\mathfrak{u}'}(X_{t+2} + R_{t+2}^{\mathfrak{u}'}(X_{t+3} + \ldots + R_{T-1}^{\mathfrak{u}'}(X_T)\ldots))),$$

then $U_t(X_{t:T}) = U_t'(X_t + R_{t,T}^{\mathfrak{u}}(X_{t+1:T}))$, $t \in \overline{\mathcal{T}} \setminus \{T\}$, and $U_t(X_T) = U_T'(X_T)$.

Proof. From the recursion, we obtain for all $t \in \mathcal{T} \setminus \{T-1\}$ that $R_{t,T}^{\mathfrak{u}}(X_{t+1:T}) = R_t^{\mathfrak{u}'}(X_{t+1} + R_{t+1,T}^{\mathfrak{u}}(X_{t+2:T}))$. Therefore,

$$\begin{aligned} U_{t+1}(X_{t+1:T}) &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R_{t,T}^{\mathfrak{u}}(X_{t+1:T})\} \\ &= \{Y \in L^\infty_{t+1} : \rho_t(Y) \le R_t^{\mathfrak{u}'}(X_{t+1} + R_{t+1,T}^{\mathfrak{u}}(X_{t+2:T}))\} \\ &= U'_{t+1}(X_{t+1} + R_{t+1,T}^{\mathfrak{u}}(X_{t+2:T})), \end{aligned}$$

which concludes the proof. □

5. Examples of Dynamic Robust Risk Measures

All dynamic uncertainty sets discussed in Section 2.3 can be used to define dynamic robust risk measures. By Proposition 2, properties of the dynamic robust risk measure follow from those of the dynamic uncertainty set. As seen in Theorem 4, if the strong time-consistency or weak recursiveness is desirable, then static uncertainty sets are called for. From Table 1, we observe that f-divergences, and in particular the KL divergence, may lead to dynamic robust risk measures that are strong t.c. whereas uncertainty sets generated by seminorms and Wasserstein distances may result in weak recursiveness. The first example illustrates a construction of dynamic robust risk measures using the dynamic uncertainty set of Example 1. Example 7 provides a dynamic robust risk measure that is strong t.c. and not normalised. Example 8 discusses dynamic robust risk measures induced by the dual representation of convex risk measures.

Example 6 (Seminorm on Random Variables). The dynamic uncertainty set $\mathfrak{u}^{\|\cdot\|}$ in Example 1, Equation (2), gives rise to the dynamic robust risk measure

$$R_t^{\mathfrak{u}^{\|\cdot\|}}(X_{t+1}) = \operatorname*{ess\,sup}_{\|X_{t+1}-Y\| \le \varepsilon_{X_t}} \rho_t(Y), \quad \forall t \in \mathcal{T}.$$

If $\|\varepsilon_{X_t}\| \le \varepsilon_{X_t}$ and $\sup\{Y : \|Y\| \le \varepsilon_{X_t}\} = \varepsilon_{X_t}$, or $u^{\|\cdot\|}$ is given by (3), then by monotonicity and translation invariance of ρ_t we obtain that

$$R_t^{\mathfrak{u}^{\|\cdot\|}}(X_{t+1}) = \operatorname{ess\,sup}\{\rho_t(Y + X_{t+1}) \in L_t^\infty : \|Y\| \le \varepsilon_{X_t}\} = \rho_t(X_{t+1}) + \varepsilon_{X_t};$$

thus, the robust risk measure additively decomposes into the risk of X_{t+1} and its uncertainty ε_{X_t}. By Theorem 4, $\mathfrak{u}^{\|\cdot\|}$ can be used to construct a weak recursive dynamic uncertainty set $\mathfrak{u}' := \{u'_t\}_{t\in\overline{\mathcal{T}}}$ through the recursive procedure $u'_T(X_T) := u_T^{\|\cdot\|}(X_T)$ and

$$u'_t(X_{t:T}) := u_t^{\|\cdot\|}(X_t + R_t^{\mathfrak{u}^{\|\cdot\|}}(X_{t+1:T}) - R_t^{\mathfrak{u}^{\|\cdot\|}}(0)), \quad t \in \overline{\mathcal{T}}/\{T\},$$

where the resulting robust risk measure is

$$R_{t,T}^{\mathfrak{u}'}(X_{t+1:T}) = R_t^{\mathfrak{u}^{\|\cdot\|}}(X_{t+1} - R_{t+1}^{\mathfrak{u}^{\|\cdot\|}}(0) + R_{t+1}^{\mathfrak{u}^{\|\cdot\|}}(X_{t+2} - R_{t+2}^{\mathfrak{u}^{\|\cdot\|}}(0) + \cdots + R_{T-1}^{\mathfrak{u}^{\|\cdot\|}}(X_T)\cdots)).$$

For the trivial norm and $0 < \varepsilon < 1$, $u^{\|\cdot\|}$ reduces to the identity. In this case, there is no uncertainty; thus, $R^{\mathfrak{u}^{\|\cdot\|}}(X_t) = \rho_t(X_t)$ and $R_{t,T}^{\mathfrak{u}'}(X_{t+1:T}) = \rho_t \circ \cdots \circ \rho_{T-1}(\sum_{i=t+1}^T X_i)$. Alternatively, if we use a p-norm, including the supremum norm, we have $R^{\mathfrak{u}'^{\|\cdot\|}}(X_t) = \rho_t(X_t) + \varepsilon_{X_t}$ and

$$R_{t,T}^{\mathfrak{u}'}(X_{t+1:T}) = \rho_t \circ \cdots \circ \rho_{T-1}\left(\sum_{i=t+1}^T X_i\right) + \rho_t \circ \cdots \circ \rho_{T-1}\left(\sum_{i=t+1}^T (\varepsilon_{X_i} - \varepsilon_0)\right). \tag{18}$$

The term $\sum_{i=t+1}^T (\varepsilon_{X_i} - \varepsilon_0)$ may represent decreasing uncertainty over time and that longer time horizons are more uncertain.

Example 7 (Strong t.c. but Not Normalised). We construct a dynamic robust risk measure that is strong t.c. but not normalised. Let the one-step risk measures be the conditional expectation, $\rho_t(\cdot) := \mathbb{E}[\cdot|\mathcal{F}_t]$, and define the time-$t$ uncertainty set

$$u_t(X_{t:T}) := \left\{ Y \in L_t^\infty : \mathbb{E}\left[Y - \sum_{i=t}^T X_i \middle| \mathcal{F}_t \right] \le \varepsilon_{t-1} \right\},$$

where $\varepsilon_{t-1} \in L_{t-1}^\infty$ is a nondegenerate random variable. The corresponding dynamic robust risk measure is

$$R_{t,T}(X_{t+1:T}) = \mathbb{E}\left[\sum_{i=t+1}^T X_i \middle| \mathcal{F}_t \right] + \varepsilon_t, \qquad t \in \mathcal{T}, \tag{19}$$

which is not normalised as $R_{t,T}(0) = \varepsilon_t \neq 0$. Moreover, R is weak recursive and further satisfies

$$R_{t,T}(X_{t+1:s} + R_{s,T}(X_{s+1:T})) = \mathbb{E}\left[\sum_{i=t+1}^T X_i \middle| \mathcal{F}_t \right] + \mathbb{E}[\varepsilon_s|\mathcal{F}_t] + \varepsilon_t.$$

From the above equations, we observe that R is strong t.c. if and only if $\mathbb{E}[\varepsilon_t|\mathcal{F}_t] = 0$, for all $t \in \mathcal{T}$. Next, we consider the normalised version of (19), that is, $\tilde{R}_{t,T}(\cdot) := R_{t,T}(\cdot) - R_{t,T}(0)$. By Lemma 4, $\tilde{R}$ is strong t.c.; however, the robustification is lost. Indeed $\tilde{R}_{t,T}(X_{t+1:T}) = \mathbb{E}[\sum_{i=t+1}^T X_i|\mathcal{F}_t]$.

Example 8 (Uncertainty Sets Induced by Dual Representation). Let $\mathcal{P}$ be the set of probability measures that are absolutely continuous w.r.t. $\mathbb{P}$ and $\{\rho_t\}_{t\in\mathcal{T}}$ a convex and Fatou-continuous sequence of one-step risk measures. Define by $\alpha_t^{min} : \mathcal{P} \to L_t^\infty \cup \{\infty\}$ the *minimal penalty functions* of $\{\rho_t\}_{t\in\mathcal{T}}$ given by

$$\alpha_t^{min}(\mathbb{Q}) := \operatorname{ess\,sup}\{\mathbb{E}_{\mathbb{Q}}[X_{t+1}|\mathcal{F}_t] : \rho_t(X_{t+1}) \le 0\},$$

where $\mathbb{E}_{\mathbb{Q}}$ denotes the expectation under $\mathbb{Q} \in \mathcal{P}$.[4] Then, the one-step risk measure has representation $\rho_t(X_{t+1}) = \operatorname{ess\,sup}_{\mathbb{Q}\in\mathcal{Q}_t} \mathbb{E}_{\mathbb{Q}}[X_{t+1} - \alpha_t^{\min}(\mathbb{Q})|\mathcal{F}_t]$, where $\mathcal{Q}_t := \{\mathbb{Q} \in \mathcal{P} : \mathbb{Q}(B) = \mathbb{P}(B) \text{ for all } B \in \mathcal{F}_t\}$ is a set of probability measures that coincide with $\mathbb{P}$ in L_t^∞. If $\mathfrak{u}$ is a static, translation-invariant, and positive homogeneous uncertainty set, then the

dynamic robust risk measure $R^{\mathfrak{u},\rho}$ admits representation

$$\begin{aligned}
R^{\mathfrak{u},\rho}_{t,T}(X_{t+1}) &= \operatorname{ess\,sup}\{\rho_t(Z) \;:\; Z \in u_{t+1}(X_{t+1})\} \\
&= \operatorname{ess\,sup}\left\{ \operatorname*{ess\,sup}_{\mathbb{Q}\in\mathcal{Q}_t}\{\mathbb{E}_{\mathbb{Q}}[Z - \alpha^{\min}_t(\mathbb{Q})|\mathcal{F}_t]\} \;:\; Z \in u_{t+1}(X_{t+1}) \right\} \\
&= \operatorname{ess\,sup}\left\{ \mathbb{E}\left[Z\frac{d\mathbb{Q}}{d\mathbb{P}} - \alpha^{\min}_t(\mathbb{Q})|\mathcal{F}_t \right] \;:\; Z \in u_{t+1}(X_{t+1}),\quad \mathbb{Q}\in\mathcal{Q}_t \right\} \\
&= \operatorname{ess\,sup}\{\mathbb{E}[Y|\mathcal{F}_t] \;:\; Y \in u^*_{t+1}(X_{t+1})\} \\
&= R^{\mathfrak{u}^*,\mathbb{E}}_{t,T}(X_{t+1}),
\end{aligned}$$

where $\mathfrak{u}^* := \{u^*_t\}_{t\in\overline{\mathcal{T}}}$ is the static uncertainty set given by

$$\begin{aligned}
u^*_{t+1}(X_{t+1}) &:= \bigcup_{\mathbb{Q}\in\mathcal{Q}_t} u_{t+1}\left(X_{t+1}\frac{d\mathbb{Q}}{d\mathbb{P}} - \alpha^{\min}_t(\mathbb{Q}) \right) \\
&= \left\{ Z\frac{d\mathbb{Q}}{d\mathbb{P}} - \alpha^{\min}_t(\mathbb{Q}) \in L^\infty_t \;:\; \mathbb{Q}\in\mathcal{Q}^\rho_t,\quad Z\in u_{t+1}(X_{t+1}) \right\}.
\end{aligned}$$

This means that any dynamic robust risk measure stemming from a family of convex one-step risk measures can be rewritten as a dynamic robust risk measure, with conditional expectations as one-step risk measures and for a suitable dynamic uncertainty set. Thus, the risk $\{\rho_t\}_{t\in\mathcal{T}}$ can be represented as the uncertainty $\mathfrak{u}^*$. However, the converse, that is, representing the uncertainty as a risk measure, is not necessarily possible, as $u_t(X_t)$ may contain $\mathcal{F}_t$-measurable random variables that are larger (or smaller) than $\sup X_t$ ($\inf X_t$).

Next, we show that any convex dynamic risk measure can be rewritten as a dynamic robust risk measure, whose one-step risk measures are the conditional expectations. For this, consider the uncertainty sets

$$u^{\mathcal{Q}_t}_t(X_t) := \left\{ X_t\frac{d\mathbb{Q}}{d\mathbb{P}} - \alpha^{min}_{t-1}(\mathbb{Q}) : \mathbb{Q}\in\mathcal{Q}_t \right\}, \qquad t\in\overline{\mathcal{T}}.$$

Then the dynamic robust risk measure coincides with the dual representation of the convex risk measure used to define the penalty function, that is,

$$\begin{aligned}
R^{u^{\mathcal{Q}_t},\mathbb{E}}_{t:T}(X_{t+1}) &= \operatorname{ess\,sup}\{\mathbb{E}[Y|\mathcal{F}_t] : Y \in u^{\mathcal{Q}_t}_{t+1}(X_{t+1})\} \\
&= \operatorname*{ess\,sup}_{\mathbb{Q}\in\mathcal{Q}_t}\{\mathbb{E}_{\mathbb{Q}}[X_{t+1}|\mathcal{F}_t] - \alpha^{\min}_t(\mathbb{Q})\} = \rho_t(X_{t+1}).
\end{aligned}$$

Appendix. Additional Proofs

The following auxiliary results are needed in the proofs that follow.

Lemma A.1. *Let ρ_t be normalised, then $A^\rho_t \subseteq A^\rho_t + A^\rho_t$. If, in addition, ρ_t is subadditive, then $A^\rho_t = A^\rho_t + A^\rho_t$.*

Proof. As ρ_t is normalized, it holds that $0 \in A^\rho_t$ and $A^\rho_t \subseteq A^\rho_t + A^\rho_t$. Next, let ρ_t be subadditive and $W \in A^\rho_t + A^\rho_t$. Then there exists $Y, Z \in A^\rho_t$ such that $W = Y + Z$. By subadditivity, $\rho_t(W) = \rho_t(Y+Z) \le \rho_t(Y) + \rho_t(Z) \le 0$, and hence, $W \in A^\rho_t$. □

Proof of Lemma 1. Let $X_{t:T}, Y_{t:T} \in L^\infty_{t,T}$ and B be an $\mathcal{F}_{t-1}$-measurable set. Item (i), assume that u_t is normalised and local. Then, using locality in the second equation and normalisation in the third, we obtain

$$u_t(1_B\, X_{t:T}) = u_t(1_B\, X_{t:T} + 1_{B^c}\, 0) = 1_B\, u_t(X_{t:T}) + 1_{B^c}\, u_t(0) = 1_B\, u_t(X_{t:T}).$$

Next, assume that $u_t(1_B\, X_{t:T}) = 1_B\, u_t(X_{t:T})$ for all $B \in \mathcal{F}_{t-1}$ and all $X_{t:T} \in L^\infty_{t,T}$. Then by taking $B = \emptyset$, u_t is normalised. We further obtain that u_t is local, by using the assumed property of U_t in the second equation

$$\begin{aligned}
u_t(1_B X_{t:T} + 1_{B^c}\, Y_{t:T}) &= 1_B\, u_t(1_B\, X_{t:T} + 1_{B^c}\, Y_{t:T}) + 1_{B^c}\, u_t(1_B\, X_{t:T} + 1_{B^c}\, Y_{t:T}) \\
&= u_t(1_B\,(1_B\, X_{t:T} + 1_{B^c}\, Y_{t:T})) + u_t(1_{B^c}\,(1_B\, X_{t:T} + 1_{B^c}\, Y_{t:T})) \\
&= u_t(1_B\, X_{t:T}) + u_t(1_{B^c}\, Y_{t:T}).
\end{aligned}$$

Item (ii), let u_t be positive homogeneous, then

$$u_t(1_B\, Z_{t:T}) = 1_B\, u_t(Z_{t:T}) + 1_{B^c}\, u_t(0), \qquad \forall Z_{t:T} \in L^\infty_{t,T}. \tag{A.1}$$

Next, we calculate, using (A.1) in the second and fourth equations, that

$$\begin{aligned} u_t(1_B X_{t:T} + 1_{B^c} Y_{t:T}) + u_t(0) &= 1_B\, u_t(1_B X_{t:T} + 1_{B^c} Y_{t:T}) + 1_{B^c}\, u_t(1_B X_{t:T} + 1_{B^c} Y_{t:T}) + u_t(0) \\ &= u_t(1_B(1_B X_{t:T} + 1_{B^c} Y_{t:T})) + u_t(1_{B^c}(1_B X_{t:T} + 1_{B^c} Y_{t:T})) \\ &= u_t(1_B\, X_{t:T}) + u_t(1_{B^c} Y_{t:T}) \\ &= 1_B u_t(X_{t:T}) + 1_{B^c} u_t(Y_{t:T}) + u_t(0). \end{aligned}$$

Subtracting $u_t(0)$ concludes that u_t is local.

Proof of Proposition 2, Items (vi)–(xiv): Item (vi), let $\{\rho_t\}_{t\in\mathcal{T}}$ and $\mathfrak{u}$ be positive homogeneous. Then, for all $0 \le \lambda \in L_t^\infty$

$$\begin{aligned} R_{t,T}(\lambda X_{t+1:T}) &= \operatorname{ess\,sup}\{\rho_t(Y) : Y \in u_{t+1}(1_{\lambda>0}\,\lambda X_{t+1:T} + 1_{\lambda=0}\,\lambda X_{t+1:T})\} \\ &= \operatorname{ess\,sup}\{\rho_t(Y) : Y \in 1_{\lambda>0}\,\lambda u_{t+1}(X_{t+1:T}) + 1_{\lambda=0} u_{t+1}(0)\} \\ &= \operatorname{ess\,sup}\{\rho_t(Y+Y') : Y \in 1_{\lambda>0}\,\lambda u_{t+1}(X_{t+1:T}), Y' \in 1_{\lambda=0} u_{t+1}(0)\} \\ &= \operatorname{ess\,sup}\left\{\rho_t(Y+Y') : 1_{\lambda>0}\frac{1}{\lambda}\, Y \in u_{t+1}(X_{t+1:T}), 1_{\lambda=0}\, Y' \in u_{t+1}(0)\right\} \\ &= \operatorname{ess\,sup}\{\rho_t(1_{\lambda>0}\lambda\, Y + 1_{\lambda=0}\, Y') : Y \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(0)\} \\ &= \operatorname{ess\,sup}\{1_{\lambda>0}\lambda\,\rho_t(Y) + 1_{\lambda=0}\,\rho_t(Y') : Y \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(0)\} \\ &= 1_{\lambda>0}\lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0} R_{t,T}(0). \end{aligned}$$

If additionally $\mathfrak{u}$ is normalized and then by Item (i) R is normalised, the above reduces to $R_{t,T}(\lambda X_{t+1:T}) = \lambda R_{t,T}(X_{t+1:T})$, and R is positive homogeneous.

Item (vii), let $\{\rho_t\}_{t\in\mathcal{T}}$ be convex and assume for $t \in \mathcal{T}$ that $u_t(\lambda\, X_{t:T} + (1-\lambda)\, Y_{t:T}) \subseteq \lambda\, u_t(X_{t:T}) + (1-\lambda)\, u_t(Y_{t:T})$ for all $\lambda \in L_t^\infty$ with $0 \le \lambda \le 1$. Next, define the $\mathcal{F}_t$-measurable random variables

$$I_0 := \begin{cases} 0 & \text{if} \quad \lambda = 0 \\ \dfrac{1}{\lambda} & \text{if} \quad \lambda > 0 \end{cases} \quad \text{and} \quad I_1 := \begin{cases} 0 & \text{if} \quad \lambda = 1 \\ \dfrac{1}{1-\lambda} & \text{if} \quad \lambda < 1. \end{cases} \tag{A.2}$$

Then, the robust risk measure satisfies

$$\begin{aligned} &R_{t,T}(\lambda\, X_{t+1:T} + (1-\lambda)\, Y_{t+1:T}) \\ &= \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(\lambda\, X_{t+1:T} + (1-\lambda)\, Y_{t+1:T})\} \\ &\le \operatorname{ess\,sup}\{\rho_t(Z) : Z \in \lambda\, u_{t+1}(X_{t+1:T}) + (1-\lambda)\, u_{t+1}(Y_{t+1:T})\} \\ &= \operatorname{ess\,sup}\{\rho_t(X'+Y') : X' \in \lambda\, u_{t+1}(X_{t+1:T}), Y' \in (1-\lambda)\, u_{t+1}(Y_{t+1:T})\} \\ &= \operatorname{ess\,sup}\{\rho_t(X'+Y') : I_0\, X' \in u_{t+1}(X_{t+1:T}), I_1\, Y' \in u_{t+1}(Y_{t+1:T})\} \\ &= \operatorname{ess\,sup}\{\rho_t(\lambda X' + (1-\lambda)Y') : X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\} \\ &\le \operatorname{ess\,sup}\{\lambda\,\rho_t(X') + (1-\lambda)\,\rho_t(Y') : X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\} \\ &= \lambda\, R_{t,T}(X_{t+1:T}) + (1-\lambda)\, R_{t,T}(Y_{t+1:T}). \end{aligned}$$

Item (viii), let $\{\rho_t\}_{t\in\mathcal{T}}$ be subadditive and assume that for all $t \in \overline{\mathcal{T}}$ and $Z \in u_t(X_{t:T} + Y_{t:T})$ there exist $X' \in u_t(X_{t:T})$ and $Y' \in u_t(Y_{t:T})$ with $Z \le X' + Y'$. Then

$$\begin{aligned} &R_{t,T}(X_{t+1:T} + Y_{t+1:T}) \\ &= \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(X_{t+1:T} + Y_{t+1:T})\} \\ &\le \operatorname{ess\,sup}\{\rho_t(Z) : Z \le X' + Y',\, X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\} \\ &= \operatorname{ess\,sup}\{\rho_t(X'+Y') : X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\} \\ &\le \operatorname{ess\,sup}\{\rho_t(X') + \rho_t(Y') : X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\} \\ &\le R_{t,T}(X_{t+1:T}) + R_{t,T}(Y_{t+1:T}), \end{aligned}$$

and $R_{t,T}$ is subadditive.

Item (xi), let $\{\rho_t\}_{t\in\mathcal{T}}$ be subadditive and assume that for all $t \in \overline{\mathcal{T}}$, it holds that $u_t(X_{t:T} + Y_{t:T}) \subseteq u_t(X_{t:T}) + u_t(Y_{t:T})$. Then

$$\begin{aligned}
&R_{t,T}(X_{t+1:T} + Y_{t+1:T})\\
&= \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(X_{t+1:T} + Y_{t+1:T})\}\\
&\le \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(X_{t+1:T}) + u_{t+1}(Y_{t+1:T})\}\\
&= \operatorname{ess\,sup}\{\rho_t(Z+W) : Z \in u_{t+1}(X_{t+1:T}), W \in u_{t+1}(Y_{t+1:T})\}\\
&\le \operatorname{ess\,sup}\{\rho_t(Z) + \rho_t(W) : Z \in u_{t+1}(X_{t+1:T}), W \in u_{t+1}(Y_{t+1:T})\}\\
&= R_{t,T}(X_{t+1:T}) + R_{t,T}(Y_{t+1:T}),
\end{aligned}$$

which shows that R is subadditive.

Item (x), the proof follows along the exact same steps as in the proof of Item (vii), with the only difference being that the inequalities are in the opposite direction.

Item (xi), let $\{\rho_t\}_{t\in\mathcal{T}}$ be superadditive and assume for $t \in \overline{\mathcal{T}}$ and $Z \in u_t(X_{t:T} + Y_{t:T})$ there exist $X' \in u_t(X_{t:T})$ and $Y' \in u_t(Y_{t:T})$ with $Z \ge X' + Y'$. Then

$$\begin{aligned}
&R_{t,T}(X_{t+1:T} + Y_{t+1:T})\\
&= \operatorname{ess\,sup}\{\rho_t(Z) : Z \in u_{t+1}(X_{t+1:T} + Y_{t+1:T})\}\\
&\ge \operatorname{ess\,sup}\{\rho_t(Z) : Z \ge X' + Y', X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\}\\
&= \operatorname{ess\,sup}\{\rho_t(X' + Y') : X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\}\\
&\ge \operatorname{ess\,sup}\{\rho_t(X') + \rho_t(Y') : X' \in u_{t+1}(X_{t+1:T}), Y' \in u_{t+1}(Y_{t+1:T})\}\\
&= R_{t,T}(X_{t+1:T}) + R_{t,T}(Y_{t+1:T}),
\end{aligned}$$

and $R_{t,T}$ is superadditive.

Item (xii), the proof follows along the exact same steps as in the proof of Item (ix), the only difference being that the inequalities are in the opposite direction.

Item (xiii), this follows from Item (ix) and Item (xii).

Item (xiv), the proof follows along the exact same steps as in the proof of Item (vi), the only difference being that the second equation is an inequality ($\le$) and that $0 \le \lambda \le 1$. Further, recall that by definition of R, the one-step risk measure $\rho_{t,s}$ is monotone and translation invariant, and thus also local; see, for example, proposition 3.3 in Cheridito et al. [18]. □

Proof of Theorem 2, Items 6–12: Item 6, let ρ_t be positive homogeneous and $R_{t,T}$ satisfy $R_{t,T}(\lambda X_{t+1:T}) = \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0} R_{t,T}(0)$, for all $0 \le \lambda \in L^{\infty}_{t-1}$. Because ρ_t is positive homogeneous, it holds by proposition 3.6 in Cheridito et al. [18] that $\lambda X \in A^{\rho}_t$ for all $X \in A^{\rho}_t$ and $0 \le \lambda \in L^{\infty}_t$. This implies that $A^{\rho}_t = \lambda A^{\rho}_t + 1_{\lambda=0} A^{\rho}_t$ for all $0 \le \lambda \in L^{\infty}_t$. Next, we obtain, using subsequently Lemma 3 (iv), the representation of A^{ρ}_t, and finally by Lemma 3 (iv) that

$$\begin{aligned}
U_{t+1}(\lambda X_{t+1:T}) &= A^{\rho}_t + R_{t,T}(\lambda X_{t+1:T})\\
&= A^{\rho}_t + \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0} R_{t,T}(0)\\
&= \lambda (A^{\rho}_t + R_{t,T}(X_{t+1:T})) + 1_{\lambda=0} (A^{\rho}_t + R_{t,T}(0))\\
&= \lambda U_{t+1}(X_{t+1:T}) + 1_{\lambda=0} U_{t+1}(0),
\end{aligned}$$

and U_{t+1} is positive homogeneous.

Item 7, let $\{\rho_t\}_{t\in\mathcal{T}}$ and R be convex and $Y_{t:T} \in L^{\infty}_{t,T}$. We need to show that for all $\lambda \in L^{\infty}_{t-1}$ with $0 \le \lambda \le 1$, that

$$U_t(\lambda X_{t:T} + (1-\lambda) Y_{t:T}) \subseteq \lambda U_t(X_{t:T}) + (1-\lambda) U_t(Y_{t:T}), \tag{A.3}$$

which by Lemma 3 (iv) is equivalent to

$$\begin{aligned}
A^{\rho}_t + R_{t,T}(\lambda X_{t+1:T} + (1-\lambda) Y_{t+1:T}) &\subseteq \lambda(A^{\rho}_t + R_{t,T}(X_{t+1:T})) + (1-\lambda)(A^{\rho}_t + R_{t,T}(Y_{t+1:T}))\\
&= A^{\rho}_t + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T}),
\end{aligned}$$

where the equality follows because if ρ_t is convex, then A^{ρ}_t is a convex set; hence, it holds that $A^{\rho}_t = \lambda A^{\rho}_t + (1-\lambda) A^{\rho}_t$. To show the inclusion, let $Z \in A^{\rho}_t + R_{t,T}(\lambda X_{t+1:T} + (1-\lambda) Y_{t+1:T})$; this means there exists a $Z' \in L^{\infty}_{t+1}$ with $\rho_t(Z') \le 0$ such that $Z = Z' + R_{t,T}(\lambda X_{t+1:T} + (1-\lambda) Y_{t+1:T})$. By convexity of R we have

$$Z \le Z' + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T}).$$

Therefore, there exists a $W \geq 0$, such that

$$Z = Z' - W + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T}).$$

By monotonicity of ρ_t and $W \geq 0$, we have that $\rho_t(Z' - W) \leq \rho_t(Z') \leq 0$, which implies that $Z \in A_t^\rho + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T})$, and we conclude that (A.3) holds.

Item 8, let $\{\rho_t\}_{t\in\mathcal{T}}$ and R be subadditive and $Y_{t+1:T} \in L^\infty_{t+1,T}$. We proceed similarly to the proof of Item 7. This means, using Lemma 3 (iv), that we need to show that

$$\begin{aligned} A_t^\rho + R_{t,T}(X_{t+1:T} + Y_{t+1:T}) &\subseteq A_t^\rho + R_{t,T}(X_{t+1:T}) + A_t^\rho + R_{t,T}(Y_{t+1:T}) \\ &= A_t^\rho + R_{t,T}(X_{t+1:T}) + R_{t,T}(Y_{t+1:T}), \end{aligned}$$

where the equality follows from subadditivity of ρ_t and Lemma A.1. For the inclusion, let $Z \in A_t^\rho + R_{t,T}(X_{t+1:T} + Y_{t+1:T})$; this means there exists a $Z' \in L^\infty_{t+1}$ with $\rho_t(Z') \leq 0$ such that $Z = Z' + R_{t,T}(X_{t+1:T} + Y_{t+1:T})$. By subadditivity of R we have

$$Z \leq Z + R_{t,T}(X_{t+1:T}) + R_{t,T}(Y_{t+1:T}),$$

which by monotonicity of ρ_t implies that $Z \in A_t^\rho + R_{t,T}(X_{t+1:T}) + R_{t,T}(Y_{t+1:T})$.

Item 9, let $\{\rho_t\}_{t\in\mathcal{T}}$ be additive, R concave, $Y_{t+1:T} \in L^\infty_{t+1,T}$, and $\lambda \in L^\infty_t$ with $0 \leq \lambda \leq 1$. We proceed similarly to the proof of Item 7. Thus, we need to show that

$$A_t^\rho + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T}) \subseteq A_t^\rho + R_{t,T}(\lambda X_{t+1:T} + (1-\lambda) Y_{t+1:T}).$$

For the inclusion, let $Z \in A_t^\rho + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T})$; this means there exists a $Z' \in L^\infty_{t+1}$ with $\rho_t(Z') \leq 0$ such that $Z = Z' + A_t^\rho + \lambda R_{t,T}(X_{t+1:T}) + (1-\lambda) R_{t,T}(Y_{t+1:T})$. By subadditivity of R we have

$$Z \leq Z + R_{t,T}(\lambda X_{t+1:T} + (1-\lambda) Y_{t+1:T}),$$

which by monotonicity of ρ_t implies that $Z \in A_t^\rho + R_{t,T}(\lambda X_{t+1:T} + (1-\lambda) Y_{t+1:T})$.

Item 10, the proof follows using similar steps as in the proof of Item 9.

Item 11 is a consequence of Items 8 and 10.

Item 12 follows the same steps of the proof of Item 6, where the second equality becomes a set inclusion, that is, $A_t^\rho + R_{t,T}(\lambda X_{t+1:T}) \subseteq A_t^\rho + \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0} R_{t,T}(0)$. This follows from $R_{t,T}(\lambda X_{t+1:T}) \leq \lambda R_{t,T}(X_{t+1:T}) + 1_{\lambda=0} R_{t,T}(0)$ and monotonicity of the one-step risk measure ρ_t. □

Endnotes

[1] A set $u \subset L^\infty_t$ is bounded if $\inf\{c \in \mathbb{R} : \mathbb{P}(c \geq |X|) = 1,\ \forall X \in u\} < \infty$, and bounded from above if $\inf\{c \in \mathbb{R} : \mathbb{P}(c \geq X) = 1,\ \forall X \in u\} < \infty$.

[2] Recall that $u_t(X_{t:T}) + Z_t := \{Y + Z_t \in L^\infty_t : Y \in u_t(X_{t:T})\}$.

[3] This theorem also holds if strong t.c. is replaced by order t.c. (see Definition 7 for details), because these notions of time-consistencies are equivalent when the dynamic risk measure is normalised, monotone, and translation invariant (see also Figure 1).

[4] An example is the conditional entropic risk measure with the (scaled) conditional KL divergence as penalty function, that is, $\rho_t(X_{t+1}) = \frac{1}{\beta}\log \mathbb{E}[\exp(\beta X_{t+1})|\mathcal{F}_t]$ and $\alpha_t^{min}(\mathbb{Q}) = \frac{1}{\beta}\mathbb{E}_{\mathbb{Q}}\left[\log\frac{d\mathbb{Q}}{d\mathbb{P}}|\mathcal{F}_t\right]$, where $\mathbb{Q} \in \mathcal{P}$ and $\beta > 0$.

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